THE CONVEX UNIFORM 5-POLYTOPES: A COMPUTER-ASSISTED CLASSIFICATION

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Abstract. A convex polytope is uniform if its symmetries act transitively on its vertices and its facets are uniform, down to regular polygons. We give a computer-assisted completeness proof for the known convex uniform 5-polytopes: Wythoff constructions of types A_5, B_5 and D_5, prisms over convex uniform 4-polytopes, and products of regular polygons with convex uniform polyhedra. The theorem holds conditional on the cited 3-D and Conway–Guy–Möller 4-D classifications, the correctness of the written reductions in Sections 2–5, and the correctness of the certificate checkers. For faces with at most 16 sides, enumeration covers all possible vertex neighborhoods, identifies 628 known polytopes and rules out one additional candidate; a face with at least 17 sides forces a product. This extends Möller's method to dimension five and produces no new examples.

AI production. The proofs, the Lean formalization and the text were produced by AI models (from OpenAI and Anthropic), working under the author's direction. The automated checks described below have been run; no human expert review of the written reductions has taken place.
Scope of verification. This is a computer-assisted classification. The Lean project checks a conditional implication and selected geometric lemmas; the complete theorem is not formalized in Lean. The computational hypotheses and their connection to the certificates remain outside Lean. This work has not been expert-reviewed. Section 6 gives the replay and Section 7 states the formalization's scope.

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1. Introduction

1.1. **Uniform polytopes.** All polytopes are convex. A *polygon* is uniform if it is regular. For $d \geq 3$, a d-polytope P is *uniform* if its symmetry group $\mathrm{Sym}(P)$ acts transitively on its vertices and each of its facets is a uniform $(d-1)$-polytope. Then every face of P is uniform (faces of facets are faces), and all edges have the same length: the edges of a uniform polytope have one length by induction on the dimension, two facets sharing a ridge share its edges, and the facet graph (facets adjacent through ridges) is connected. We always scale to *unit edges*. Uniform 3-polytopes are the uniform polyhedra, uniform 4-polytopes are also called uniform polychora, and uniform 5-polytopes uniform polytera. The Lean predicates `Uniform5.Composition.Uniform` and `UnitUniform` in `lean/Uniform5/CompositionGeometry.lean` use this same geometric definition and unit-edge normalization.

Notation used throughout:

- $\{p\}$ is the regular p-gon with unit edges, of circumradius $r_p = 1/(2\sin(\pi/p))$; the distance between the two neighbors of a vertex of $\{p\}$ is $\mathrm{ch}_p := 2\cos(\pi/p)$.
- $P_q = \{q\} \times [0,1]$ is the q-gonal prism (P_4 is the cube) and A_q the q-gonal antiprism (A_3 is the octahedron), with unit edges. The height of A_q is $h_A(q)$ with $h_A(q)^2 = 1 - \frac{1}{4\cos^2(\pi/(2q))}$, and its circumradius is R_{A_q} with $R_{A_q}^2 = r_q^2 + h_A(q)^2/4$.

- For polytopes $X \subset \mathbb{R}^a$, $Y \subset \mathbb{R}^b$ the product $X \times Y \subset \mathbb{R}^{a+b}$ is again a polytope; $\{p\} \times \{q\}$ is a *duoprism*, $K \times \{\} := K \times [0,1]$ a *prism*. A product of uniform polytopes with unit edges is uniform with unit edges.
- Sym, Stab, aff, conv, relint have their usual meanings; VF denotes a vertex figure (Section 2).

Definition 1.1 (Wythoffian polytopes [1, 2])**.** Let W be a finite irreducible Coxeter group of rank k acting on $\mathbb{R}^k$, with simple roots $\alpha_0, \ldots, \alpha_{k-1}$ (nodes of the Coxeter diagram), and let $\emptyset \neq J \subseteq \{0, \ldots, k-1\}$ (the *ringed* nodes). Let x_J be the point at distance $1/2$ from the mirrors of the ringed nodes and on the mirrors of the others, on the positive side of all of them. The *Wythoffian polytope* $W_J = \mathrm{conv}(Wx_J)$ is uniform with unit edges. We name it by the group and the ring pattern, e.g. B_5 `00001`; node orders are those of Table 2 in Appendix B.

The two non-Wythoffian non-prismatic convex uniform 4-polytopes are the *snub 24-cell* (the 600-cell minus the vertices of an inscribed 24-cell) and the *grand antiprism* (the 600-cell minus two great decagons in orthogonal planes) [8].

1.2. **The theorem.** This paper presents a completeness statement with explicit computational lemmas and reproduction commands. It is a specialist classification result about the known five-dimensional list, not a classification in arbitrary dimension or a new general method. It is a computer-assisted result conditional on the cited 3-D and 4-D classifications (Facts 1.4 and 1.5); the separate Lean composition checks an implication from named inputs, with the limits stated in Section 7.

Main Theorem. *Assume Facts 1.4 and 1.5. Every convex uniform 5-polytope is congruent, after scaling to unit edges, to one of the following:*

(K1) *a Wythoffian polytope W_J of one of the Coxeter groups A_5, B_5, D_5 (19 distinct A_5 forms, 31 B_5 forms and 8 further D_5 forms);*

(K2) *a prism $K \times \{\}$ over a convex uniform 4-polytope K; 45 Wythoffian bases, plus the snub 24-cell and grand antiprism, give 47 prisms, of which 46 are not already in (K1), since the tesseract prism is the 5-cube;*

(K3) *a product $\{p\} \times Q$ of a regular polygon, $p \geq 3$, with a convex uniform polyhedron Q: a Platonic or Archimedean solid, a prism P_q (giving the* duoprism prisms *$\{p\} \times \{q\} \times \{\}$), or an antiprism A_q.*

The descriptions overlap (the 5-cube is a B_5 Wythoffian, the prism over the tesseract and $\{4\}\times$cube; prisms over polyhedral prisms, antiprism prisms and duoprisms are of the form (K3)). The theorem asserts coverage, not disjointness. Counting the Coxeter–Johnson constructions [2, 9, 15] with the four-dimensional bases of [18] gives $19 + 31 + 8 + (47 - 1) = 104$ in this convention: the tesseract prism is counted once as the 5-cube. The older Open Problem Garden total of 105 [19] has not been reconciled with that convention; it is not used by the proof. The proof instead generates every ring pattern and product and tests congruence (Section 3). Every polytope in (K1)–(K3) is convex and uniform, so the theorem is a classification. It is the union of two theorems, proved in Sections 4 and 5:

Theorem 1.2 (Case A)**.** *Every convex uniform 5-polytope all of whose 2-faces have at most 16 sides is congruent to one of the 628 distinct polytopes of the list $\mathcal{K}_{16}$ of Section 3 (all members of (K1)–(K3) whose polygons have at most 16 sides).*

Theorem 1.3 (Case B)**.** *Every convex uniform 5-polytope with a 2-face of at least 17 sides is congruent to $\{n\} \times Q$ with Q a convex uniform polyhedron, or to $\{q\} \times A_n$, for integers $n \geq 17$ and $q \geq 3$.*

Both conclusions of Theorem 1.3 are of type (K3). So Theorems 1.2 and 1.3 give the Main Theorem.

1.3. **History and status.** Gosset [12] enumerated the semiregular polytopes (regular facets) in all dimensions; G. Blind and R. Blind [6] proved that list complete. Elte's 1912 monograph [11] is another early treatment of higher-dimensional semiregular polytopes. Coxeter developed the Wythoff construction and the theory of uniform polytopes [9, 10], and Johnson's thesis [15] set up the general theory. The convex uniform polyhedra are the Platonic and Archimedean solids, prisms and antiprisms; the list of all uniform polyhedra, convex or not, was shown complete by Sopov [23] and Skilling [22]. In dimension four Conway and Guy announced in 1965 a computer-assisted completeness proof, finding one new non-Wythoffian polytope, the grand antiprism [8]; a complete account of a computer-assisted proof is Möller's thesis [17, 18]: 1000 edge neighborhoods, each with a forced circumradius, and 64 vertex neighborhoods, each matched with a known polychoron, with the prism and antiprism families treated by hand.

The present proof builds directly on Möller's classification and its edge-neighborhood, forced-radius and vertex-neighborhood enumeration strategy. The published three-dimensional classification supplies the possible polyhedral faces, and Conway–Guy and Möller supply the possible four-dimensional facets. Coxeter's and Johnson's constructions supply the known families whose completeness is at issue here. These foundational classifications and constructions are credited inputs; the contribution of this paper is the conditional five-dimensional completeness argument and its checked computations.

Schulte's 2017 Handbook chapter states that "a complete list is known only for $d = 4$" [21]. Winter's Appendix E.3 states: "it is unknown whether there exist non-prismatic non-Wythoffian uniform polytopes in dimension five or above." [24, p. 167] The Open Problem Garden and community catalogs also record the question as open [19, 20]. These sources support the status of the problem; they are not completeness premises.

Johnson's 1966 thesis [15] develops constructions in arbitrary dimension, including dimension five; its discussion of known enumeration (pp. 226, 291–296) does not establish completeness of the convex five-dimensional list. Johnson's unpublished 1991 manuscript [16] is cited by Schulte for the four-dimensional list; its full text is unavailable, so no five-dimensional completeness claim is attributed to it here. Bowers's and Klitzing's catalogs [3, 4] document known examples and nomenclature; Bowers's search includes nonconvex polytera and is not presented here as an exhaustive search of convex examples. Neither catalog is a completeness premise, and the construction lists used here are generated from Wythoff and Cartesian-product recipes.

Related verification design. The verification design was developed with Càllisto's honeycomb work [7] as a known methodological precedent, including its division between Lean arguments and external finite certificate checks; none of its theorems or certificates is a premise here. The geometric enumeration method used here is Möller's.

1.4. **Dependence on the 3- and 4-dimensional classifications.** The proof uses two classifications as cited inputs. They enter only through the following two facts.

Fact 1.4 (3-D classification [21, Section 18.5], [13])**.** *A convex uniform polyhedron is a Platonic solid, one of the 13 Archimedean solids, a prism P_q ($q \geq 3$) or an antiprism A_q ($q \geq 4$). It is determined up to congruence by its vertex configuration (the cyclic sequence of polygon sizes around a vertex). In particular a convex uniform polyhedron with a q-gonal face, $q \notin \{3, 4, 5, 6, 8, 10\}$, is P_q or A_q.*

Fact 1.5 (4-D classification [8, 17])**.** *A convex uniform 4-polytope is one of: the Wythoffians of A_4, B_4, D_4, F_4, H_4; the snub 24-cell; the grand antiprism; a prism $Q \times \{\}$ over a convex uniform polyhedron Q; a duoprism $\{p\} \times \{q\}$. In particular a convex uniform 4-polytope with a q-gonal 2-face, $q \notin \{3, 4, 5, 6, 8, 10\}$, is a duoprism $\{q\} \times \{p\}$ (including $P_q \times \{\} = \{q\} \times \{4\}$) or the antiprism prism $A_q \times \{\}$.*

The vertex-transitivity in our definition matters: the pseudo-rhombicuboctahedron (elongated square gyrobicupola) has congruent vertex configurations but is not vertex-transitive, and is not uniform [13]. Fact 1.5 is itself computer-assisted. As a check of the method, the certified shared pipeline in dimension four recovers all 152 known objects at family cutoff 14, including the grand antiprism and snub 24-cell (Section 6.3). This bounded positive control is not a link of the 5-D proof or a proof of the cited all-parameter 4-D classification.

1.5. **Strategy.** Let P be a convex uniform 5-polytope with unit edges and circumradius R. Section 2 proves the reduction: P is inscribed in a sphere; around every 2-face G the facets form a cycle, and R is a root of an explicit *closure equation* of that cycle; at a vertex v the neighbors of v span the vertex figure VF, a 4-polytope inscribed in a 3-sphere whose facets are the vertex figures of the facets of P at v; two distinct vertices are at least one edge apart (the separation lemma), which lets interval enclosures decide vertex identity when they are narrow enough (and report it as undecided otherwise); and the star of one vertex, if it forces the stars of its neighbors, forces P.

Case A (Section 4): all polygons have at most 16 sides. Then every facet belongs to a finite library of 181 congruence classes of uniform 4-polytopes. Stage 1 enumerates every cycle of library facets around a polygon and certifies all roots of its closure equation (14,965 cycles); radii that can occur together form 563 *radius classes* after a combinatorial filter. Stage 2 enumerates candidate signatures covering every actual vertex figure in each class (629 in total). Stage 3 covers every actual star by retained encodings, certifies the scalar-root assertion, and proves identification and forcing conditional on realization. For 628 encodings, any represented actual star is identified with that of a separately constructed known polytope Q and forces $P \cong Q$. The one extra candidate is excluded from every convex uniform polytope by a development that gets stuck; no exact local realization of that candidate is asserted.

Here **181 counts four-dimensional facet types**, not five-dimensional polytopes. At the same polygon cutoff the known 5-D list has 890 names representing 628 congruence classes (Section 3); without the cutoff it includes infinite families. The facet library is checked by exact construction and incidence checks and by a separate rebuild in `checks/independent-s1-b2/` (Computation 3.5). Neither rebuild proves Möller's classification anew.

Case B (Section 5): some polygon F has $n \geq 17$ sides. Around F only three cycle types are possible, each forcing R by a formula in n and finitely many parameters. A certified computation (B2) shows that this R is never a root of any Case-A cycle, so every polygon at a vertex lies in a facet containing a large polygon. A written structural argument, with two more certified computations ((P1) and (W)), then forces P to be a product.

Why the cutoff is 16/17. Seventeen is a sufficient, not an optimal, threshold: it exceeds every exceptional polygon size in the lower-dimensional classifications and satisfies $2\sin(2\pi/17) < 1$ (Lemma 5.4). The B2, P1 and W certificates cover the entire remaining range. The small-face radius cover has $\max \Omega = 63.2195709614\ldots$; it includes closure roots that need not extend to any polytope, so this is not the largest radius of a known polytope with at most 16-sided faces. B2 checks $17 \leq n \leq 397$ and the corresponding auxiliary parameters, then uses $r_{398} > 63.344 > \max \Omega$ and monotonicity for the tails (Section 5.3).

The computations state the finite or interval claims they check and the written arguments needed to interpret them. Section 6.2 records shared dependencies; Section 7 states the formal scope.

What independence means here. Separate implementations share mathematical constructions, some input lists, model certificates and arithmetic libraries. The exact-exclusion checks also share the stage-1 construction code. P1's sampled tails are non-proof checks; its proof uses written angle, chord and monotonicity arguments. Section 6.2 specifies these dependencies.

Why this does not yet classify dimension six. In dimension six, the facets would be the five-dimensional families proved complete here, but completeness of that input list alone does not supply a finite search. The codimension-three faces are now polyhedra rather than polygons; their

neighborhoods require new incidence and metric data, and the five-dimensional vertex figures make assembly and forcing larger. Most importantly, the large-polygon fan argument of Case B does not classify all unbounded families in dimension six. A new parameter reduction and certified bounds would be needed before a finite replay could prove completeness there.

2. The reduction

Throughout this section P is a convex uniform d-polytope with unit edges, $d \geq 3$ (we use $d = 5$ from Section 2.4 on), with vertex set V.

2.1. **Inscribed sphere and face centers.**

Lemma 2.1. *Let c be the centroid of V. Then all vertices lie on the sphere of some radius R about c, and c lies in the interior of P. For every face Φ of P the orthogonal projection o_Φ of c onto* $\operatorname{aff}\Phi$ *is the circumcenter of Φ, and lies in the relative interior of Φ. If Φ has circumradius R_Φ, then $|c - o_\Phi| = \sqrt{R^2 - R_\Phi^2}$.*

Proof. Every symmetry permutes V and so fixes c; by vertex-transitivity all vertices are equidistant from c. The centroid of the vertices of a full-dimensional polytope is an interior point. The vertices of Φ lie on the intersection of the sphere with $\operatorname{aff}\Phi$, a sphere in $\operatorname{aff}\Phi$ about o_Φ; they affinely span $\operatorname{aff}\Phi$, so o_Φ is the unique circumcenter of Φ. Since Φ is uniform, applying the first part to Φ shows that its circumcenter is its vertex centroid, which is relatively interior. The last formula is Pythagoras. □

2.2. **The separation lemma.**

Lemma 2.2 (Separation lemma)**.** *Let X be a convex polytope inscribed in a sphere, with at least three vertices, let v be a vertex and w a vertex nearest to v among the other vertices. Then $[v, w]$ is an edge of X. Consequently, in a convex uniform polytope with unit edges any two distinct vertices are at distance at least* 1*.*

Proof. Let B be the closed ball with diameter $[v, w]$, center m and radius $\rho = |v - w|/2$. If a vertex $x \notin \{v, w\}$ lay in B, the angle $\angle vxw$ would be at least 90°, so $|v-w|^2 \geq |v-x|^2+|x-w|^2 > |v-x|^2$, contradicting the choice of w. Let S be the circumsphere, with center c and radius R, and consider the difference of powers $f(x) = (|x - m|^2 - \rho^2) - (|x - c|^2 - R^2)$, an affine function. It vanishes at v and w, and at every other vertex x we have $|x - c| = R$ and $|x - m| > \rho$, so $f(x) > 0$. In particular f is not constant, so $H = \{f = 0\}$ is a hyperplane, $X \subseteq \{f \geq 0\}$, and $X \cap H$ is the face $\operatorname{conv}\{v, w\}$. So $[v, w]$ is an edge. If all edges have length 1, the nearest other vertex of any vertex is at distance 1, so all distances between distinct vertices are at least 1. □

Corollary 2.3 (Decision rule)**.** *Let x, y be two vertices of one convex uniform polytope with unit edges (possibly hypothetical), and let $[\ell, u]$ be a rigorous enclosure of $|x - y|^2$. Then:*

(a) *if $\ell \leq 0$ and $u < 1$, then $x = y$;*
(b) *if $0 < \ell$ and $u < 1$, we have a contradiction (the configuration does not occur);*
(c) *if $0 < \ell$ (and $u \geq 1$), then $x \neq y$;*
(d) *if $\ell \leq 0$ and $u \geq 1$, nothing is concluded: $x = y$ and $x \neq y$ both remain possible.*

Case (d) is *undecided*. A search that meets it must either refine the enclosure, keep both possibilities, or stop and report the branch as open; treating the two points as distinct (or as equal) would not be conservative. Every search used by the proof handles (d) as undecided, and the full replays recorded below met it zero times (Sections 4.2, 4.5.2 and 4.5.3): `cert_vf.py` (`identify`) raises `Undecided`, counts it, and refuses to return a complete class; `check_u2` raises `PrecisionError` and exits nonzero; `checks/check_u3` uses the stricter equality bound $u < 1/4$ and turns undecided cases into `OPEN`, never into a refutation.

2.3. **Vertex figures.**

Lemma 2.4. *Let P be a convex uniform d-polytope with unit edges and circumradius R, centered at 0, and v a vertex. The neighbors of v (the other ends of the edges at v) are exactly the vertices at distance 1 from v. They lie on the hyperplane $H = \{x : x \cdot v = R^2 - \frac{1}{2}\}$, every other vertex except v lies strictly on the side $x \cdot v < R^2 - \frac{1}{2}$, and* $\mathrm{VF} := H \cap P = \mathrm{conv}(\text{neighbors of } v)$ *is a $(d-1)$-polytope inscribed in a sphere of radius $r = \sqrt{1 - 1/(4R^2)}$ about $o_{\mathrm{VF}} = (1 - \frac{1}{2R^2})v$. The faces of* VF *correspond to the faces of P containing v (a j-face $\Phi \ni v$ gives the $(j-1)$-face $H \cap \Phi$). For a facet $K \ni v$ of circumradius R_K,* $\mathrm{VF}(K) := H \cap K$ *is inscribed in a sphere of radius $r_K = \sqrt{1 - 1/(4R_K^2)}$ in a hyperplane of H at distance $\sqrt{r^2 - r_K^2}$ from o_{VF}. An edge of* VF *coming from a p-gon has length $\mathrm{ch}_p = 2\cos(\pi/p)$. The point o_{VF} lies in the relative interior of* VF.

Proof. By Lemma 2.2 the vertices at distance 1 from v are neighbors, and conversely neighbors are at distance 1. For a vertex x, $|x-v|^2 = 2R^2 - 2x \cdot v$, so $|x-v| = 1$ iff $x \cdot v = R^2 - \frac{1}{2}$, and $|x-v| > 1$ (every other vertex $\neq v$) iff $x \cdot v < R^2 - \frac{1}{2}$. Hence H separates v from all non-neighbors, and $H \cap P$ is the convex hull of the neighbors; this is the standard vertex figure, whose face lattice is that of the faces through v. A neighbor x makes an angle θ with v at the center with $2R\sin(\theta/2) = 1$, so its distance from the axis $\mathbb{R}v$ is $R\sin\theta = \cos(\theta/2) = \sqrt{1 - 1/(4R^2)}$. The same computation inside K gives r_K; the circumcenter of $\mathrm{VF}(K)$ is the projection of o_{VF}, which gives the distance. In a regular p-gon the two neighbors of a vertex are ch_p apart. Finally, write $o_{\mathrm{VF}} = t_0 v$ with $t_0 = 1 - 1/(2R^2)$. Since 0 is the centroid of the N vertices, $q := -v/(N-1)$ is the centroid of the vertices other than v, so $q \in P$, and $q \cdot v \le R^2 - \frac{1}{2}$, since every other vertex x has $x \cdot v \le R^2 - \frac{1}{2}$, with equality exactly for the neighbors. If some vertex other than v is not a neighbor, the inequality is strict, i.e. $-1/(N-1) < t_0$; also $t_0 < 1$. Then o_{VF} lies on the open segment from q to v, which passes through the interior point 0; every point of this open segment is interior to P (it lies strictly between the interior point 0 and a point of P, or is 0). Hence o_{VF} is an interior point of P lying in H, i.e. a relative interior point of $\mathrm{VF} = H \cap P$. If every vertex other than v is a neighbor (as in the 5-simplex), then VF is the convex hull of all other vertices, $t_0 = -1/(N-1)$ and $o_{\mathrm{VF}} = q$ is the centroid of the vertices of VF, which is a relative interior point of VF. □

Remark 2.5. The center of the vertex figure need not lie on the segment from v to the center of P: when $R < 1/\sqrt{2}$, $t_0 < 0$; for the 5-simplex, $R^2 = 5/12$ and $t_0 = -1/5$. The centroid argument above handles both cases, including 4-simplex facets when applied in Lemma 4.10.

2.4. **Cycles around a polygon and the closure equation.**

From now on $d = 5$. Let G be a 2-face of P, a p-gon with circumcenter o_G and circumradius $\rho = r_p$, and let N be the 3-dimensional orthogonal complement of the direction space of $\mathrm{aff}\, G$. The faces of P containing G are cells (3-faces) and facets; modulo $\mathrm{aff}\, G$ the tangent cone of P at G is a pointed polyhedral 3-cone in N, whose extreme rays correspond to the cells $C \supset G$ and whose 2-faces correspond to the facets $K \supset G$. Going around the cone gives a cyclic sequence $(K_1, C_1, K_2, C_2, \ldots, K_k, C_k)$, $k \ge 3$, where facet K_i contains the cells C_{i-1} and C_i (indices mod k). Let $u_C \in N$ be the unit vector of the ray of C and α_i the dihedral angle of the 4-polytope K_i at its 2-face G, between the cells C_{i-1} and C_i; this is the angle between $u_{C_{i-1}}$ and u_{C_i}. We call $(p; K_1, C_1, \ldots, K_k, C_k)$ the *cycle* of G; cycles are taken up to rotation and reversal.

Lemma 2.6 (Closure equation)**.** *With this notation, let $w = c - o_G \in N$, $h = |w| = \sqrt{R^2 - \rho^2}$, and for each cell $C \supset G$ of circumradius R_C let $d_C = \sqrt{R_C^2 - \rho^2} > 0$. Then:*

(a) $\sum_i \alpha_i < 2\pi$;
(b) $w \cdot u_C = d_C$, *so the angle δ_C between w and u_C satisfies* $\cos\delta_C = d_C/h$;

(c) *w lies strictly inside the cone spanned by the u_C, and with $\theta_i \in (0,\pi)$ the angle at $\hat{w} = w/h$ of the spherical triangle $(\hat{w}, u_{C_{i-1}}, u_{C_i})$,*

$$\cos\theta_i = \frac{\cos\alpha_i - \cos\delta_{C_{i-1}}\cos\delta_{C_i}}{\sin\delta_{C_{i-1}}\sin\delta_{C_i}}, \qquad \sum_{i=1}^{k}\theta_i = 2\pi;$$

(d) *$R > R_K$ for every facet $K \supset G$.*

Proof. (a) The intersection of a pointed polyhedral 3-cone with the unit sphere is a convex spherical polygon in an open hemisphere, whose side lengths are the face angles α_i; its perimeter is less than 2π. (b) By Lemma 2.1 applied to C, o_C is the projection of c on $\operatorname{aff} C$ and o_G the projection of o_C on $\operatorname{aff} G$; so $o_C - o_G \in N$ points along the ray of C (as $o_C \in \operatorname{relint} C$), has length $d_C > 0$, and $w - (o_C - o_G)$ is orthogonal to $\operatorname{aff} C$, hence to u_C. (c) The tangent cone of P at G is the sum of the direction space of $\operatorname{aff} G$ and a pointed cone $T_N \subset N$, and the interior of P lies in o_G plus its interior; since c is interior and $w = c - o_G \in N$, w lies in the interior of T_N, the cone spanned by the u_C. The triangles $(\hat{w}, u_{C_{i-1}}, u_{C_i})$ are then non-degenerate, their angles at $\hat{w}$ fill the full angle around $\hat{w}$, and the formula is the spherical law of cosines for the side α_i opposite $\hat{w}$. (d) $|c - o_K| = \sqrt{R^2 - R_K^2}$ is the distance from the interior point c to the hyperplane of K, which is positive. □

In the variable $s = 1/R$ the cosines become $\cos\delta_C(s) = d_C\, s/\sqrt{1-\rho^2 s^2}$, increasing in s, and the *closure function* of the cycle is

$$g(s) = \sum_{i=1}^{k}\theta_i(s) - 2\pi, \qquad 0 \le s \le 1/R_{\max},$$

where $R_{\max} = \max_i R_{K_i}$ and $s = 0$ stands for $R = \infty$.

Remark 2.7. On this domain $\delta_{C_{i-1}} + \delta_{C_i} \ge \alpha_i$, so the argument of the arccosine is at least -1: at $s = 1/R_{K_i}$ the point $\hat{w}$ is the direction of $o_{K_i} - o_G$, which lies on the arc from $u_{C_{i-1}}$ to u_{C_i} (equality), and the δ's decrease in s. If instead $|\delta_{C_{i-1}} - \delta_{C_i}| > \alpha_i$ the triangle does not exist; on such s the configuration is impossible. The implementations clip the lower end at -1 (sound by this remark) and discard boxes on which a triangle is certified impossible.

Why the closure equation is not an identity. For an admissible cycle, $\theta_i(0) = \alpha_i$ and hence $g(0) = \sum_i \alpha_i - 2\pi < 0$ by Lemma 2.6(a). Thus the closure equation cannot hold identically on its radius domain. For the finite library we have the stronger, checked monotonicity that also rules out an accumulation of roots at an endpoint. Put $a = d_A^2$, $b = d_B^2$, $D = R_K^2 - \rho^2$ and $c = \cos\alpha$ for a word, and let $u = 1/(R^2 - \rho^2)$. Its closure-angle cosine is

$$f(u) = \frac{c - \sqrt{ab}\,u}{\sqrt{(1-au)(1-bu)}}.$$

Since $D > a, b > 0$, the sign of $f'(u)$ on $[0, 1/D]$ is the sign of

$$-\sqrt{ab} + \tfrac{1}{2}c(a+b) + u\big(\tfrac{1}{2}\sqrt{ab}(a+b) - cab\big).$$

This is affine in u. The independent stage-1 certificate checks strict negativity at both endpoints for all 558 library words; hence every closure angle, and their sum, increases strictly in u on the nondegenerate domain. Each Case-A cycle has at most one admissible finite root (Computation 4.1). A flat cycle has $g(0) = 0$ but its closure function is still not identically zero: the same strict monotonicity makes the closure sum exceed 2π for $u > 0$. Flat cycles are excluded in any event by convexity. For Case B, Section 5.3 gives the explicit radius formulas and finite root equations. The finite-radius conclusion therefore uses more than an appeal to analyticity alone.

Two kinds of cycles satisfy the closure equation only spuriously and are excluded exactly:

Lemma 2.8 (Exact exclusions)**.** (Flat.) *A cycle with $\sum \alpha_i = 2\pi$ exactly is not the cycle of a 2-face of a convex 5-polytope.* (Degenerate.) *A root at $R = R_{\max}$ exactly does not give a convex uniform polytope.*

Proof. Lemma 2.6(a) and (d). □

For a flat cycle, $g(0) = 0$, the formal infinite-radius limit. Degenerate roots put the center on a facet hyperplane. Interval arithmetic cannot certify these equalities; they are proved exactly (Computations 4.1 and 3.6). A local angle-sum identity alone does not give a global honeycomb.

2.5. **Faces, simple vertices.**

Lemma 2.9 (Face lemma)**.** *Let X be a polytope, Φ a face of X and p, q vertices of Φ. If $[p, q]$ is an edge of X, it is an edge of Φ. Conversely every edge of Φ is an edge of X.*

Proof. $[p, q] \subseteq \Phi$ by convexity, and $[p, q] = [p, q] \cap \Phi$ is an intersection of two faces of X, hence a face of X contained in Φ, hence a face of Φ. The converse holds because a face of a face is a face. □

Lemma 2.10 (Simple vertices)**.** *A vertex x of degree 4 in a 4-polytope is simple: each facet containing x contains exactly three of the four edges at x, and each 3-subset of the edges at x is the set of edges at x of exactly one facet.*

Proof. The vertex figure of x is a 3-polytope with four vertices, a tetrahedron; its facets (all four triangles) correspond to the facets through x. □

2.6. **Stars and local uniqueness.** The *star* of a vertex v is the set of facets of P containing v, each with its full vertex set.

Lemma 2.11 (Local uniqueness gives global uniqueness)**.** *Let S be the star of v_0 in some convex uniform 5-polytope, and suppose: for every convex uniform 5-polytope P with star S at v_0 and every neighbor w of v_0, the star of P at w is one and the same set Σ_w, independent of P. Then any two convex uniform 5-polytopes P_1, P_2 with star S at v_0 are equal.*

Proof. We show by induction on the edge distance from v_0 that P_1 and P_2 have the same vertices and the same stars there. Suppose P_1 and P_2 have the same star Σ at a vertex w. By vertex-transitivity of P_1 there is an isometry g_1 with $g_1(v_0) = w$ and $g_1(S) = \Sigma$. Then $g_1^{-1}(P_1)$ and $g_1^{-1}(P_2)$ are convex uniform 5-polytopes with star S at v_0; by hypothesis they have the same star at every neighbor u' of v_0. Applying g_1, P_1 and P_2 have the same star at every neighbor $u = g_1(u')$ of w (the neighbors of w are read off the common star Σ). The edge graph of a polytope is connected (Balinski [5]), and a polytope is the convex hull of its vertices, so $P_1 = P_2$. □

The induction is written out here because a symmetry of P_1 mapping v_0 to w need not be a symmetry of P_2; the hypothesis is applied to $g_1^{-1}(P_i)$, not to P_i.

2.7. **The case split.** A 2-face is *large* if it has at least 17 sides. *Case A*: P has no large 2-face. *Case B*: P has a large 2-face. The cases are disjoint and exhaustive. Polygon sizes $7, 9, 11, \ldots, 16$ (which occur only in prisms, antiprisms and duoprisms) belong to Case A, together with the sizes $3, 4, 5, 6, 8, 10$ of the non-prismatic polyhedra and 4-polytopes.

3. The finite library and the known list

3.1. **The facet library.** For $3 \le P \le 16$ let $\mathcal{L}_P$ be the following list of names of convex uniform 4-polytopes all of whose 2-faces have at most P sides. For $P = 16$ it has 214 names:

- the Wythoffians `A4`*abcd*, `B4`*abcd*, `D4`*abcd*, `F4`*abcd*, `H4`*abcd* (*abcd* $\in \{0,1\}^4 \setminus \{0000\}$; 75 names; their 2-faces have at most 10 sides);
- `sadi` (snub 24-cell) and `gap` (grand antiprism);

- `pr-`X, the prism over X, for the 18 Platonic and Archimedean solids X;
- `antiprism-prism-`$n = A_n \times \{\}$, $3 \le n \le 16$ (14 names);
- `duoprism-`p`-`$q = \{p\} \times \{q\}$, $3 \le p \le q \le 16$ (105 names; the prisms $P_q \times \{\}$ are `duoprism-4-`q).

Names, diagrams and cell labels are listed in Appendix B. For smaller P, the definition of $\mathcal{L}_P$ retains exactly the names in this list whose every 2-face has at most P sides.
The small control. The $P = 6$ quick control $\mathcal{H}_6$ retains fixed 8- and 10-gon facets and has 136 classes rather than the literal cutoff's 129, whereas $\mathcal{H}_{10} = \mathcal{L}_{10}$ and $\mathcal{H}_{16} = \mathcal{L}_{16}$; we write P_{ctl} for its family-parameter bound.

Lemma 3.1 (Library completeness)**.** *Every convex uniform 4-polytope all of whose 2-faces have at most 16 sides is congruent to the polytope of some name in $\mathcal{L}_{16}$.*

Proof. Fact 1.5: the Wythoffians and the two snub-type polytopes have only 3-, 4-, 5-, 6-, 8- and 10-gons; the prisms over polyhedra X with at most 16-gonal faces are the `pr-`X, the duoprisms `duoprism-4-`q and the antiprism prisms (Fact 1.4); and the duoprisms are listed for $p, q \le 16$. □

Several names denote congruent polytopes (e.g. `A40001` and `A41000` are both the 5-cell, and `B40001`, `pr-cube` and `duoprism-4-4` are the tesseract). The enumeration keeps one *canonical* name per congruence class: 181 classes, with 33 aliases (Table 3). The canonical types are 45 Wythoffians, the two snub-type polytopes, 17 polyhedral prisms, 13 antiprism prisms and 104 duoprisms. A wrong merge would silently drop a facet type, so the merges are proved exactly.

Certified computation 3.2 (MERGE: exact congruence of the 33 aliases)**.** Each of the 33 aliases is congruent to its canonical name.

What is checked. `merge.py` builds both vertex sets exactly in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ (elements stored as four rationals in the basis $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$): Wythoffians from explicit simple roots, first checked exactly against the Coxeter diagram; the point x_J by an exact rational Gram solve, rechecked; the orbit by exact breadth-first search; the products in closed form. A floating-point search proposes a bijection; it is accepted only if the vertex counts agree and every pairwise squared distance is preserved exactly. By polarization the image vectors then have the same Gram matrix, so the bijection extends to an affine isometry of the affine hulls. The program also checks that its alias list equals the alias list of the shared facet file `checks/check_u1/inputs/u1_facets_p16.json` (option `--facets`). Result: 33/33 congruences proved; four corruptions rejected (two false pairs, one vertex moved by 10^{-12}, two images swapped). *Second implementation.* `checks/independent-merge-filter/` independently derives the Wythoff and product coordinates in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ and certifies all 33 aliases by vertex bijections preserving 465,206 unordered squared distances. A separate affine checker agrees; both reject seven semantic corruptions. The alias list is a shared input; the later library reconstructions still take it as input.

3.2. **Facet data.** For every library type K the computations need: its circumradius R_K; for each orbit of 2-faces under $\mathrm{Sym}(K)$, the polygon size p, the two cells at the polygon and the dihedral angle α between them; and the *model* of K: a base vertex $K[0]$, its neighbors $K[1], \ldots$ (the vertex figure X_K), its full vertex set, and the list of 2-faces of X_K (the cells at $K[0]$) with their cyclic orders and cell labels. Every polygon orbit has a member through $K[0]$ by vertex-transitivity, so all of this is local data at one vertex.

Certified computation 3.3 (LIB: facet data)**.** For each of the 181 canonical types, the circumradius and the dihedral angles of all polygon orbits are enclosed in balls, and the model is constructed with all coordinates enclosed in balls.

What is computed. `cert_facets4.py`, `cert_star.py`, `cert_full.py` (python-flint/Arb, 200 bits): Wythoffians from rigorous Wythoff coordinates (`cert_wythoff.py`); prisms and antiprism prisms from the vertex-configuration data of the polyhedra (`cert_poly3.py`, below); duoprisms in closed

form; the snub 24-cell and the grand antiprism from exact $\mathbb{Q}(\sqrt{5})$ coordinates of the 600-cell; full vertex sets as orbits under exact reflections or (for the snub cube and snub dodecahedron) under their rotation groups, moved into the model frame by isometries verified on balls. `cert_poly3.py` computes a uniform polyhedron's circumradius and edge angles from its vertex configuration alone: the neighbors of a vertex lie on a circle, the central angles of the chords ch_{q_i} sum to 2π, and the unique solution is enclosed. Exact values: `exact_angles.py` computes $\cos\alpha$ exactly for every non-snub facet angle ($\cos^2\alpha$ lies in $\mathbb{Q}(\sqrt{d})$, $d \in \{1, 2, 5\}$, computed in the root basis; exact $\mathbb{Q}(\sqrt{5})$ coordinates for `sadi` and `gap`; closed forms for products), and all agree with the balls.

The following certificate supplies a certified correspondence from cells and labels to every polygon-orbit and angle entry used by stage 1, including the word-angle property.

Lemma 3.4 (A word determines its angle)**.** *Let G be a p-gon of a uniform 4-polytope K (circumradius R_K), between cells A, B of circumradii R_A, R_B, and let $\rho = r_p$. Then the dihedral angle of K at G is*

$$\alpha = \arccos\sqrt{\frac{R_A^2 - \rho^2}{R_K^2 - \rho^2}} + \arccos\sqrt{\frac{R_B^2 - \rho^2}{R_K^2 - \rho^2}}.$$

In particular the word (K, p, A, B) determines α, even when its polygon occurrences lie in different orbits of $\mathrm{Sym}(K)$.

Proof. In the plane through the center f of G perpendicular to G, the two cells give rays u_A, u_B bounding a convex wedge of angle α, and the center O of K projects strictly inside it (Lemma 2.1 applied to K, and Lemma 2.6(b),(c) one dimension down). With $D = |O - f|^2 = R_K^2 - \rho^2$ we have $(O - f) \cdot u_A = \sqrt{R_A^2 - \rho^2} > 0$ and likewise for B, since the projection of O on each cell is its relatively interior circumcenter. The direction of $O - f$ lies between the rays, so the two angles to it add to α. □

Certified computation 3.5 (LINK: certified stage-1 input)**.** For each of the 181 canonical types (assuming the construction properties of Section 3.3), the list of polygon occurrences at the base vertex, with their two cells and labels, is complete and correct (1,277 occurrences, 558 distinct words (K, p, A, B)); the exported circumradius encloses R_K and each exported angle encloses the angle of Lemma 3.4. Every radius and angle entry of the stage-1 input tables at $P_{\mathrm{ctl}} = 6$, 10 and 16 contains these rigorous enclosures, and each table lists exactly the words of its corresponding control library $\mathcal{H}_P$.

What is checked. Wythoff neighbors are generated in exact $\mathbb{Q}$, $\mathbb{Q}(\sqrt{2})$ or $\mathbb{Q}(\sqrt{5})$ coordinates; `sadi` and `gap` from all 120 vertices of the 600-cell in exact $\mathbb{Q}(\sqrt{5})$ arithmetic; prisms, antiprism prisms and duoprisms from their exact product identities (which replace the small-value zero tests that `checks/check_u1` uses for these types). Floating-point hulls only propose faces; every face then passes the ridge certificate (Computation 4.8: exact coplanarity or the product identity, strict supporting signs, strict cyclic order, chord labels, two faces per edge), and closure of the face graph proves the list complete. Two separately written link checkers (one reading adjacency from the sides of the certified faces, one from pairwise face intersections and an independent dihedral computation) accept every occurrence and each rejects 15 semantic corruptions. The input-table containment check (`compare_legacy.py`, both checkers) passes at $P_{\mathrm{ctl}} = 6$ (76 radii, 243 words, 627 occurrences), 10 (106/333/815) and 16 (181/558/1,277), each rejecting six further corruptions; the 558 words are also exactly those used by the independent stage-1 checker's cycle file. *Second implementation.* The independent library of `checks/independent-s1-b2/` (Section 4.5.4), written from this paper's statements, reconstructs the 181 canonical types by exact parabolic Coxeter combinatorics, exact supporting planes for `sadi` and `gap` and product identities, and obtains the same 558 words and 1,277 occurrences; all 181 radius and 558 angle enclosures lie inside the recorded input enclosures. The two link checkers above share the ridge certificate and the constructions; the

independent library shares the alias table, the construction theorem of Section 3.3 and the written exceptional-model proofs, as well as the Arb and SymPy arithmetic libraries. In particular, `checks/independent-s1-b2/library.py`, `checks/independent-s1-b2/parabolic.py` and `checks/independent-s1-b2/exceptional.py` rebuild the geometry; `checks/independent-s1-b2/library_verify.py` checks its local incidence complex and the comparison checks containment in the consumer intervals. The construction proofs and the external 4-D classification remain common mathematical premises. The alias list is a common inventory input, though all 33 merges have separate exact congruence checks. Arithmetic-library errors and a common misinterpretation of the recursive-uniform definition are also not eliminated by matching counts. The independent rebuild reduces dependence on the original construction program; it does not make every upstream premise independent.

With Computation 3.5, the stage-1 proof uses certified input tables and the two implementations described in Section 4.1.

Exact inputs of the exclusions. LINK certifies interval enclosures. The exact exclusions of stage 1 (Lemma 2.8) need more: the algebraic cosine and radius used in each exclusion must be *exactly* the geometric quantity of the named incident word. A floating-point reconstruction or numerical recognition of a cosine does not prove this exact correspondence. By Lemma 3.4, with $\rho = r_p$, $D = R_K^2 - \rho^2$, $a = R_A^2 - \rho^2$ and $b = R_B^2 - \rho^2$, the cosine of the word's angle is

$$\cos\alpha = \frac{\sqrt{ab} - \sqrt{(D-a)(D-b)}}{D},$$

with the positive square roots fixed by the geometry ($0 < a, b < D$). So exact radii and exact cell labels determine every exact cosine.

Certified computation 3.6 (EXACT: certified exact exclusion inputs)**.** The squared circumradii of the 181 canonical types and of the 44 named cell types and, for each of the 558 incident words, the algebraic cosine used by the exclusions, are exactly the values of the constructions of Section 3.3; each cosine equals the displayed expression with positive square roots.

What is checked. The exact proposals are rebuilt by both checkers from exact constructions (the exact parabolic, quadratic-field, 600-cell and product routines of `checks/independent-s1-b2/`); neither accepts the file by digest or by numerical agreement. Checker A requires the exact residual $Dc^2 - 2c\sqrt{ab} + a + b - D = 0$, decided by an exact minimal-polynomial test, together with $\sqrt{ab} - Dc > 0$, strict positivity of a, b, $D - a$, $D - b$ by outward arithmetic, and an exact residual for every radius against its construction. Checker B squares once more, verifies $4c^2ab = (Dc^2 + a + b - D)^2$, restores the sign, authenticates the radii by its own exact Gram solves and product identities (for the snub factors, a Sturm count isolating the unique admissible root of the eliminated polynomial), projects the incident word labels from cell intersections, and decides zero by exact number-field embedding. Neither accepts an interval containing zero as an equality; `nsimplify` and rational-angle guesses are only proposals and need an exact zero residual. The live stage-1 facet table is then authenticated against these values: complete names, aliases and words, containment of every exact radius and angle after the stage-1 parser, and the depth bound $9\alpha > 2\pi$. `exact_angles.py` takes its operands only from this certified source (`exact_inputs.py`). The flat identities are proved using positive sines, the exact product $\prod(c_j + i\,s_j) = 1$, and a certified sum in $(\pi, 3\pi)$; the degenerate exclusions use only that the three-ray Gram matrix is nonsingular and the opposite Cramer numerator is exactly zero, which already rules out a strictly interior center. The consumer entry point `authenticate_exclusions` rechecks every requested deletion on the actual cycle's words and exact operands and rejects duplicate, overlapping, unproved or out-of-range indices. *Result.* Both checkers accept 181 facet radii, 44 cell radii and 558 cosines; a separate direct sweep of the normal-cosine residual passes all 558 words. The exclusions at the three bounds are 790/1,703/7,080 flat cycles and 23/39/57 degenerate roots at $P_{\mathrm{ctl}} = 6/10/16$.

Corruptions. Twelve test methods pass, rejecting in both checkers perturbations of a facet radius, a cell radius or a cosine by 10^{-40}, the wrong square-root branch, a lost sign, wrong labels, polygons and multiplicities, missing and duplicate words and facets, floating operands, false flat and degenerate claims (four near-right angles, a 4π winding, a singular Gram matrix), malformed exclusion lists and root records, and $\sqrt{2}+10^{-40}$ replaced by $\sqrt{2}$. *Shared input and second implementation.* Both `check_inputs.py` and `check_inputs_b.py` import `exact_source.py`. That bridge calls `checks/independent-s1-b2/library.py` and `checks/independent-s1-b2/library_verify.py`, with the parabolic, exceptional and product constructions behind them. This is one shared construction implementation, not two independent library reconstructions. Checker B supplies its own radius solve in `radii_b.py` and incidence projection; it does not thereby make the common constructions independent. A common error in that construction layer could survive both input checks; their agreement alone does not rule it out. The exclusions themselves are proved twice: by `exact_angles.py` with the independent certificate check `check_exclusions_b.py`, and by the separately written stage 1 of `checks/independent-s1-b2/`, which finds the same 7,080 flat cycles and 57 degenerate roots with its own exact Gram matrices and exact phase products (it uses the same construction code).

3.3. **Model constructions.** The computations use explicit constructions ("models") of the library facets and of the known 5-polytopes. It is not enough that these are *named* correctly: the proof uses that each constructed point set is the vertex set of a uniform polytope with unit edges, with the stated facet types and complete vertex orbits, since the separation lemma is applied to model vertices. These properties are supplied as follows (`checks/model_audit/`).

- *Wythoff constructions, antiprisms and products* (`checks/model_audit/wythoff_proof.md`): a written proof, from standard facts on finite Coxeter groups, of the supporting-face theorem, unit edges and recursive uniformity of W_J for every binary ring pattern of the groups used, of the product and antiprism constructions, of the facet rule and of the named aliases; and an explicit contract under which an indexed interval orbit enumeration returns exactly the orbit, index by index.
- *Snub 24-cell and grand antiprism*: full convex-hull uniformity of the 96- and 100-vertex subsets of the 600-cell and containment of the actual indexed coordinates, checked by two independent implementations (rational-pair and number-field), each rejecting eight corruptions.
- *Snub cube and snub dodecahedron*: complete algebraic constructions with exact generator and index identities and coordinate containment, checked independently with interval/Krawczyk and rational-dyadic arithmetic, each rejecting nine corruptions.
- *Runtime guard*: `model_guard.py` enforces in every run the midpoint, width and grid conditions of that contract for every point-set and lookup query, using exact rational endpoints. It is installed before any model is built in the independent stage-2 and stage-3 drivers, the equivalence checkers and the template auditor, and kept through their searches; a guard failure is reported as `OPEN`, never as an exclusion. All 181 facet models (79,445 vertices; 273,802 point-set and 2,066 lookup queries) pass, and every final stage-2 and stage-3 run below records the guard active with zero failures.

Conventional names matter only for reporting. The two independent packages of Section 4.5.4 build their own models in exact number fields (Coxeter orbits by exact reflections without rounding, products and antiprisms from exact recurrences, the snub factors from isolated algebraic roots, the two 600-cell deletions from the 120 exact points, recovered from certificate data and checked against them in `checks/independent-s23/`). Both rely on the same written construction theorem and exceptional-model proofs `checks/model_audit/snub_PROOF.md` and `checks/model_audit/q5_PROOF.md`; `checks/independent-s23/` also uses the two exceptional model certificates.

3.4. **The known list.** For $P \leq 16$, $\mathcal{K}_P$ is the list of names

- `A5`*abcde*, `B5`*abcde*, `D5`*abcde*, all 31 ring patterns each (93 names);

- `prism-`K for the canonical types K of $\mathcal{L}_P$;
- $\{p\} \times X$ for $3 \le p \le P$ and X a Platonic or Archimedean solid, P_q or A_q, all of whose faces have at most P sides.

For $P = 16$ this gives $93 + 181 + 14 \cdot 44 = 890$ names. They denote 628 pairwise non-congruent polytopes: the stage-3 identification (Section 4.5.3) assigns every one of the 890 names to one of 628 vertex-figure signatures with verdict `KNOWN`. Names with the same signature are congruent, because actual stars with that signature have the same rooted encoding and radius, and their star forces the polytope (Proposition 4.14); existence comes from the named constructions, not from retaining a search row; names with different signatures are not, because the signature of a vertex figure is a congruence invariant. This count is not taken from the literature. By construction $\mathcal{K}_{16}$ contains every polytope of (K1)–(K3) whose 2-faces have at most 16 sides: the Wythoffians have only 3-, 4-, 5-, 6-, 8- and 10-gons, a prism over a prismatic 4-polytope is of the form (K3), and every (K2) prism over a non-prismatic 4-polytope is a `prism-`K.

4. Case A: every 2-face has at most 16 sides

In this section P is a convex uniform 5-polytope with unit edges, circumradius R, center c, all of whose 2-faces have at most 16 sides. By Lemma 3.1 every facet of P is congruent to a canonical type of $\mathcal{L}_{16}$, and every cell is a uniform polyhedron with at most 16-gonal faces (Fact 1.4). We write $P_{\max} = 16$; the same pipeline was also run with $P_{\mathrm{ctl}} = 6$ and 10 as controls, with the counts of Table 1.

4.1. **Stage 1: cycles and their radii.** A *library cycle* is a cycle $(p; K_1, C_1, \dots, K_k, C_k)$ with $p \le 16$, $k \ge 3$, each K_i a canonical type having a p-gon orbit between cells of types C_{i-1}, C_i (dihedral angle α_i), and $\sum \alpha_i$ not certified to exceed 2π. By Lemma 2.6, the cycle of every 2-face of P is a library cycle. By Lemma 3.4 the word (K, p, A, B) of a polygon orbit determines its dihedral angle, so the word $(p; K_1, C_1, \dots, K_k, C_k)$ determines the angles and the closure function; stage 2 and stage 3 treat cycles as such words.

Certified computation 4.1 (S1: stage 1)**.**

(i) The list $\mathcal{C}_{16}$ of library cycles, up to rotation and reversal, has 14,965 elements, and it is complete.

(ii) For each cycle the zeros of its closure function g on $[0, 1/R_{\max}]$ (with the domain extended to the upper end of the ball enclosing $1/R_{\max}$) lie in a listed finite set of *root intervals*, formed by merging retained leaves of width below 10^{-20} in s; this threshold does not bound the width of a merged interval. Of the cycles, 14,428 have at least one root interval, 537 have none, and none is undecided.

(iii) Exactly 7,080 cycles have $\sum \alpha_i = 2\pi$ exactly (flat), and exactly 57 root intervals are the root $R = R_{\max}$ exactly (degenerate); each is proved by exact algebra.

What is computed. `cert_stage1_5d.py` (Arb, 160–200 bits) enumerates cycles as closed walks in the multigraph whose vertices are cell types and whose edges are the p-gon orbits of the facet types, pruning when the certified angle sum exceeds 2π. The smallest facet angle in the library is $\pi/3$ (in $\{3\} \times \{q\}$ at its q-gons, the interior angle of the triangle), so a non-flat cycle has at most 5 facets and a library cycle at most 6 (six angles $\pi/3$ are flat); the search goes to 8. On $[0, 1/R_{\max}]$ the closure function is evaluated on boxes by ball arithmetic, using the monotonicity of $\cos \delta_C(s)$ for the range of each δ_C and the clipping of Remark 2.7; a box is discarded when the enclosure of g excludes 0 or a triangle is certified impossible, and the remaining boxes are bisected below width 10^{-20} and merged into root intervals. Adjacent retained leaves can produce a merged interval wider than 10^{-20}; no upper bound of that size is needed for coverage or the later whole-class root check. Root intervals are written as outward-rounded 45-digit decimal bounds with a margin of 10^{-32}; the merge of adjacent boxes is exact. `exact_angles.py` proves flatness, on the exact cosines of Computation 3.6 only: angles that are rational multiples of π are verified as such exactly (a

rational-angle guess is accepted only after an exact cosine identity), the others are combined into the claim $\exp(i\sum\alpha_j)=1$, which is verified by an exact minimal-polynomial zero test (sympy), and the ball of $\sum\alpha_j$ lies in $(\pi,3\pi)$; so $\sum\alpha_j=2\pi$. The output records 30/31/32 proved cosine tuples at $P_{\mathrm{ctl}}=6/10/16$, including $2\arccos(-\sqrt{6}/4)+\arccos(-1/4)=2\pi$, and $\arccos(-\sqrt{1/3-2\sqrt{5}/15})+\arccos(-\sqrt{1/3+2\sqrt{5}/15})+\arccos(-1/\sqrt{5})=2\pi$ (pentagonal antiprism, icosidodecahedron and dodecahedron angles). All degenerate roots occur in 3-cycles. For a 3-cycle the link is a rigid spherical triangle and the closure has one explicit solution: with Γ the Gram matrix of the three cell rays ($\Gamma_{ij}=\cos\alpha$) and $d_i=\sqrt{R_{C_i}^2-\rho^2}$, the unnormalized center offset is $w=\sum\lambda_i u_i$ with $\lambda=\Gamma^{-1}d$ (for the unit direction $\hat{w}$ multiply by the positive scale $1/|w|$); the center lies on the hyperplane of facet K iff the coefficient of the ray opposite K vanishes, a Cramer determinant, which is proved exactly zero. Since Γ is nonsingular, the center offset is unique, and an exactly zero coefficient already excludes a strictly interior center, whatever the signs of the other two; the exclusion uses nothing more. A positive rescaling of the center does not change the vanishing determinant.

Proof route. The stage-1 run uses input tables certified by Computation 3.5, with the exclusions on the exact inputs of Computation 3.6. *Second implementation.* The stage 1 of `checks/independent-s1-b2/` (Section 4.5.4) enumerates all closed walks of length three to six on its own library (every angle is at least $\pi/3$, so longer walks exceed 2π and length six is flat), and isolates roots using a strict monotonicity proved for each word: in the variable $u=1/(R^2-\rho^2)$ the derivative of the cosine of each closure angle has the sign of an affine function of u, certified strictly negative at both ends of the admissible interval, so each angle increases strictly in u (decreases in R) and every non-flat cycle has at most one root. It reproduces the 14,965 cycles with identical words and statuses, the 7,080 flat cycles, 57 degenerate roots and 537 cycles without roots, and 7,291 retained roots, each contained in the recorded interval of the same cycle; its root intervals are narrower. *Shared with the proof route:* the alias table, the construction theorem and, for the exclusions, the exact construction code of Computation 3.6; both use Arb (python-flint) and SymPy.

Definition 4.2. Ω is the set of root intervals of all cycles in $\mathcal{C}_{16}$ other than the flat cycles and the degenerate roots: 7,291 intervals, all bounded, with $\min\Omega=0.6454972\ldots$ (the 5-simplex) and $\max\Omega=63.2195709614\ldots$.

The export (`export_omega.py`, function `project_intervals`, shared by both loaders) removes a flat cycle as a whole, as Lemma 2.8 allows, but Lemma 2.8 excludes only the degenerate *root*, not the whole cycle. The export therefore removes a cycle carrying a degenerate root only after checking that it has exactly one finite, positive certified root interval, and raises otherwise. All 23/39/57 degenerate cycles at $P_{\mathrm{ctl}}=6/10/16$ pass this check; the exports contain 1,629/3,516/7,291 intervals, and an added second root on an actual excluded cycle is rejected by both loaders at each P_{ctl}. The export entry point validates the records structurally before exporting, regenerates the complete cycle inventory against the actual facet table, and calls the exact authentication of Computation 3.6 on the actual excluded cycles, requiring the returned index sets to equal the requested ones; `project_intervals` implements the structural step. Nine entry-point controls (malformed authentication results and failing guards) are rejected at all three bounds.

Corollary 4.3. *Let X be any convex uniform 5-polytope with unit edges and circumradius R_X, and G a 2-face of X all of whose facets (around G) belong to $\mathcal{L}_{16}$, with G at most 16-gonal. Then R_X lies in a root interval of the cycle of G, and that interval belongs to Ω.*

Proof. The cycle of G is a library cycle (Lemma 2.6(a)), hence in $\mathcal{C}_{16}$; it is not flat (Lemma 2.8); $1/R_X$ is a zero of g in $[0,1/R_{\max})$ by Lemma 2.6(c),(d), so it lies in a root interval, which is not a degenerate one (Lemma 2.8). □

The facets around G need not be in $\mathcal{L}_{16}$ when X has larger polygons elsewhere; Corollary 4.3 is stated for any X because Case B uses it.

Radius classes and the closure filter. A *radius class* is a connected component of the overlap graph of the intervals of Ω (closed intervals that meet are joined); its hull is the class interval. A class carries the set of cycles with an interval in it. The *closure filter* then removes, repeatedly until nothing changes, every cycle of a class containing a facet type K such that some polygon orbit (p, C_a, C_b) of K is not a side (K between C_a and C_b at a p-gon) of any remaining cycle of the class. A class *survives* if cycles remain.

Lemma 4.4. *All cycles of the 2-faces of P at a vertex v lie in one radius class, the class containing R, and none of them is removed by the closure filter.*

Proof. By Corollary 4.3 each of these cycles has a root interval in Ω containing R, so all these intervals lie in the class of R. Let K be a facet of P at v and (p, C_a, C_b) a polygon orbit of K. Since K is vertex-transitive, the orbit has a member $G \ni v$, a 2-face of P whose cycle contains K between C_a and C_b; this cycle is in the class of R. So the set of cycles of P's 2-faces at v is never touched by the filter (by induction over the iterations). □

Certified computation 4.5 (FILTER)**.** The intervals of Ω form 4,224 radius classes, of which 563 survive the closure filter.

The program `class_filter.py` applies this combinatorial filter to the stage-1 output. The program rejects unresolved, unknown, missing or malformed statuses, malformed or non-finite surviving intervals, and invalid, duplicate or overlapping exclusion lists; it reconstructs the complete canonical cycle inventory from the facet table and compares it with the input (so a dropped record of either status is detected); it compares endpoints and forms class hulls in exact decimal arithmetic; and before forming classes it runs both LINK checkers on the actual table and the exact authentication of Computation 3.6. Rerun on the pinned stage-1 roots with regenerated exact exclusions, it gives 760/1,790/4,224 radius classes and 136/278/563 surviving classes at $P_{\mathrm{ctl}} = 6/10/16$, with all class files byte-identical to their pins; 23 controls pass. *Second implementation.* `checks/independent-merge-filter/` independently computes exact rational interval components and fixed-point deletions, with explicit missing-word witnesses. Its separate certificate checker imports no search code; both checkers reject the same three semantic corruptions. Complete surviving class cycle sets match the $\mathcal{H}_6$, $\mathcal{H}_{10}$ and $\mathcal{H}_{16}$ controls (136/278/563); the literal $\mathcal{L}_6$ result is 129 (Section 3.1). The certified stage-1 intervals, statuses, facet words and aliases are supplied inputs, not re-proved by FILTER. The stage-1 checker `checks/check_u1` also finds the same 4,224 classes and 563 survivors from its own roots, but its inputs carry the uncertified decisions listed in Section 6.2. The independent stage-2 and stage-3 programs of `checks/independent-s23/` take these classes as input and do not recompute them.

4.2. **Stage 2: vertex figures.** Let I be a surviving class with interval $[R_{\mathrm{lo}}, R_{\mathrm{hi}}]$. By Lemmas 2.4 and 4.4, if $R \in I$ then the vertex figure VF of P at v satisfies:

(a) VF is a convex 4-polytope inscribed in the 3-sphere of radius $r = \sqrt{1 - 1/(4R^2)}$; each facet is VF(K) for a facet type K that occurs in a cycle of I, lying in a hyperplane at distance $\sqrt{r^2 - r_K^2}$ from the center, with the center on its inner side;
(b) every ridge (2-face) of VF is the vertex figure of a cell and lies in exactly two facets of VF, with the same cell label;
(c) around every edge of VF (a polygon of P at v) the facets and ridge labels, read cyclically, form a cycle of I;
(d) distinct vertices of VF are at distance at least 1 (Lemma 2.2).

A *signature* of a vertex figure is its multiset of facet types together with the multiset of canonical cyclic words around its edges.

Certified computation 4.6 (S2: stage 2)**.** For each of the 563 surviving classes, an exhaustive search, sound for every R in the class interval, outputs a finite list of complete candidate encodings; their signatures are 629 in total (one class, the class at $R = 1.0505014948\ldots$ formed by a cycle of three 5-cells and a 16-cell around a triangle, has none). For every R in the class and every vertex figure satisfying (a)–(d), its signature is one of the signatures listed for the class. This is a coverage statement: retaining an encoding does not prove that all its assembly equations have a simultaneous exact solution. Existence of the known models is supplied separately by Section 3.3 and their named constructions.

Why such a search is exhaustive. Fix the least facet type t_0 present: the first facet is X_{t_0} in a fixed position, which is no loss of generality after an isometry. Repeatedly take an open ridge (a ridge in only one placed facet). In the true vertex figure exactly one other facet contains it (by (b)). Trying every facet type $\geq t_0$, every ridge of that type with the same label and size, every cyclic and reversed correspondence of the ridge polygon, and both positions of the hinge (the two hyperplanes through the ridge at the right distance from the center; only the one on the side opposite the existing facet can occur, by step 2 of the proof of Lemma 4.10) therefore includes the true facet. A branch may be cut only for a certified violation of (a)–(d): a vertex certified strictly outside a facet hyperplane; two vertices at a certified distance in $(0, 1)$; three facets at a ridge or a label clash; a closed cycle around a VF-edge that is not a cycle of the class. These rejection rules are exhaustive. Rounded or approximately symmetric equivalences are insufficient for this purpose. The proof route below removes candidates only by *exact* equivalences.

Lemma 4.7 (Exact equivalence of placements)**.** *Let g be an exact isometry of the model vertex figure X_K fixing its center, preserving its labeled ridges and VF-edge polygon sizes, and mapping an ordered source ridge correspondence o to another one o' (that is, $g(o_i) = o'_i$). Then for every admissible R in the class ($R > R_K$) the facets placed over a given target ridge by o and by o', with the hinge on the side opposite to the existing facet, coincide as labeled point sets.*

Proof. g fixes the template center, so it preserves the distance of the source ridge from it and carries its outward normal to that of o'. With the hinge on the opposite side, the new facet's normal is computed from $d_K(R)$, that distance and the target ridge alone, so it is the same for both options. If A_o, $A_{o'}$ are the two gluing maps, $A_{o'} \circ g$ and A_o have the same ordered ridge images and the same outward normal; they agree on three non-collinear ridge vertices and on the facet center, which is not in their affine plane, and so they are equal on the facet hyperplane. Hence the placed facets coincide as functions of R, whether or not g reverses orientation, and g preserves the adjacent labels used by later filters. □

Proof route (`check_u2` with `stage2`; see). The independent stage-2 search (Section 4.5.2) builds every VF(K) directly from the guarded exact construction models of Section 3.3 (`exact_types.py`): metric constants by directed interval operations, cell lists, cyclic orders and labels from the model-backed cell certificates, no inverse cosine. The only reductions are exact stabilizer permutations of the construction model, restricted to the neighbors of the base vertex and checked to preserve labeled ridges and polygon sizes; a subgroup suffices, since fewer symmetries only retain more options. The *equivalence certificate* `equivalences-p16.json` lists, for each of the 181 types, every ordered ridge correspondence (cyclic and reversed): 5,108 in total, 1,437 retained and 3,671 discarded, each discarded one with an explicit exact permutation carrying it to a retained one, so that Lemma 4.7 applies. Two checkers (`audit_certificates.py` and the separately written `check_equivalences.py`, which imports neither the search nor the certificate producer) check ordered correspondence, labels, edge types, the hinge flag, full coverage of the options, exact group membership and that every representative is retained; each accepts all 5,108 correspondences and rejects all 905 deliberate corruptions (five per type). Vertex identity uses the stricter bound $u < 1/4$ and an undecided distance raises `PrecisionError`, so the class is reported undecided and the run exits nonzero (a

control widens a radius interval deliberately and checks this). *Result.* On the certified stage-1 class intervals the guarded searches complete every class with zero undecided, error or non-strict outputs: $P_{\mathrm{ctl}} = 6$: 136 classes, 168 signatures, 2,081 nodes; $P_{\max} = 10$: 278, 332, 3,556; $P_{\max} = 16$: 563, 629, 5,869 nodes (no node cap). The same 629 signatures are found on the independent stage-1 checker's broader class intervals. Both certificate checkers accept all 5,108 correspondences and reject all 905 corruptions. They share template, interval and model-guard code, so they check one certificate without providing two independent proofs of the model library. The replay driver `replay_all.py` requires zero non-strict completions, checks every per-figure strict flag and the complete class accounting, and rejects 18 corruptions of its evidence; the full strict replays at $P_{\mathrm{ctl}} = 6, 10, 16$ cover all 977 classes.

Second implementation (`checks/independent-s23/`, Section 4.5.4). The implementation uses exact number-field models, Arb at 320 bits, the opposite hinge of Lemma 4.10, its own exact equivalence certificate (all 181 models and 5,108 ordered options, checked by a separate checker), cuts only for certified separation, supporting planes, ridge multiplicity, labels, polygon sizes or a partial edge link that extends no allowed cycle, and no search cap (finiteness from a packing bound). On the same class intervals it completes 136/278/563 classes for $\mathcal{H}_6$, $\mathcal{H}_{10}$ and $\mathcal{H}_{16}$ with 168/332/629 signatures, equal to the proof route's, with no undecided class. *Shared with the proof route:* the stage-1 classes, the facet-name table and alias table, the construction theorem, the exceptional model certificates and the separation lemma.

4.3. **Model ridge lists.**

Certified computation 4.8 (RIDGES: certified ridge lists)**.** For each of the 181 canonical types K, the list of 2-faces of the model vertex figure X_K (which point sets, their cyclic order, their cell labels) is correct.

What is checked (`cert_ridges.py`), ridge by ridge:

(1) *Face.* A plane through three consecutive points of the ridge; every other point of X_K certified strictly on one side; the ridge's remaining points on the plane *exactly*: for Wythoffian types the neighbors of $K[0]$ are recomputed in exact arithmetic ($\mathbb{Q}$ for A_4, D_4; $\mathbb{Q}(\sqrt{2})$ for B_4, F_4; $\mathbb{Q}(\sqrt{5})$ for H_4), matched to the model points by the separation lemma inside K, and the differences have exact affine rank 2; `sadi` and `gap` likewise in $\mathbb{Q}(\sqrt{5})$; the polyhedral base ridge of a prism or antiprism prism lies in a plane by construction (its points and $K[0]$ have last coordinate exactly $1/2$, and lie on two spheres); all other ridges are triangles.
(2) *Cyclic order.* For consecutive points a, b every other ridge point is certified strictly on one side of the line ab in the ridge plane; so consecutive pairs are sides, hence edges of X_K.
(3) *Completeness.* Every side lies in exactly two listed ridges and every point in a listed ridge; since the face graph of a 3-polytope is connected, the list contains every 2-face.
(4) *Label.* A side of the ridge of a cell is the vertex figure of a q-gon, of length ch_q; q is pinned by the certified bracket $4\cos^2(\pi/(q-1)) < |a-b|^2 < 4\cos^2(\pi/(q+1))$ (the function $4\cos^2(\pi/q)$ increases with q). The cyclic sequence of sizes is the cell's vertex configuration, which determines the cell (Fact 1.4); it must match exactly one label.

Result: 181/181 types certified; 823 ridges (747 triangles, 55 exactly coplanar by exact arithmetic, 21 by construction), 1,277 VF-edges, each in exactly two ridges. Five wrong lists (a wrong label, a dropped ridge, a wrong cyclic order, a square split into two triangles, a non-coplanar four-point set) are rejected, and a non-coplanar set has exact affine rank 3. This certifies the local data of the models; that the model points are the vertex figure of a uniform polytope with unit edges is supplied by Section 3.3, and the correspondence of these certified cells and labels with every stage-1 input entry by Computation 3.5. *Independent replay:* the stage-3 checker certifies its own templates (cells, neighbors and symmetries) from exact models (Section 4.5.3).

4.4. **Stage 3: from vertex figures to polytopes.** Stage 3 is checked by `checks/check_u3/` (Section 4.5.3; Proposition 4.14) and `checks/independent-s23/` (Section 4.5.4). Both certify a unique radius on the whole class interval, cover every actual star by candidate encodings, identify known polytopes by elimination, and refute the remaining encoding by the development of Computation 4.12.

4.4.1. *The exact radius.*

Certified computation 4.9 (ROOT: unique radius)**.** For every one of the 629 vertex-figure signatures, one of its edge cycles has a closure function g with exactly one zero on the *whole* class interval $[1/R_{\mathrm{hi}}, 1/R_{\mathrm{lo}}]$ of the surviving stage-1 class: g has certified nonzero values of opposite signs at the two ends, and g', enclosed over the whole interval, excludes 0.

Proof route (`closure.py`, step 2 of Section 4.5.3): sign change at R_{lo} and R_{hi} and a forward-mode derivative excluding 0 on the whole exported stage-1 class interval, evaluated with certified template angles. The inverse cosine used there (`checks/check_u3/ivla.py`, `iacos`) is a proved enclosure: for an input interval $[a, b] \subseteq [-1, 1]$, an ordinary `mp.acos` evaluation only *proposes* bounds $l \le u$, and each is returned only after the outward-rounded interval cosine certifies $0 \le l \le \pi$, $\cos l \ge b$ and $0 \le u \le \pi$, $\cos u \le a$; since cosine decreases on $[0, \pi]$ this gives $l \le \arccos b \le \arccos a \le u$. A proposal that cannot be certified is replaced by the safe domain bound 0 or π, which can only make a caller report uncertainty, never a false sign. *Coverage*: the audit `audit_radius_route.py` checks, by exact decimal comparison, that every surviving stage-1 class is exported, that each exported interval contains its certified input interval and equals the interval the run used, and that every signature is accounted for with zero radius, verdict and development uncertainty. In the final guarded runs all 168/332/629 radii at $P_{\mathrm{ctl}} = 6/10/16$ are unique on their whole classes; a separate check confirms these outputs. *Second implementation:* `checks/independent-s23/` (`s3_radius.py`, Section 4.5.4) proves a unique root on the same whole class intervals for all 629 signatures, with Arb inverse cosine and its own closure constants from the exact models.

4.4.2. *Congruence from discrete data. Discrete data.* A vertex figure's data are, for each of its facets, the facet's point set, its library type K, and a correspondence c (model point $t \mapsto$ VF point $c(t)$). Correspondences are taken modulo a set E_K of *exact* isometries of X_K fixing $K[0]$: the permutations induced by generators of $\mathrm{Stab}_K(K[0])$ (exact reflections and diagram automorphisms, matched by the separation lemma inside K), and, for prisms over polyhedra, the relabellings $t \mapsto s \pm t$ of the polyhedron's vertex-figure polygon that preserve its vertex configuration (a rotation or reflection of the circle, exactly; this adds the mirror symmetry of the snub vertex figures, which Stab does not contain). Elements that do not map labeled ridges to ridges with the same label are dropped. A missing element of E_K can only make the check below fail, never pass.

Lemma 4.10 (Congruence from discrete data)**.** *Let X, X' be convex uniform 5-polytopes with the same circumradius R and vertex figures V, V' indexed by the same set, with the same facets: the same index sets, the same types, and correspondences that agree modulo E_K. Then a linear isometry A with $AV_i = V'_i$ for all i exists.*

Proof. (1) Take a facet f with $V[S_f] = g(X_K)$ and $V'[S_f] = g'(X_K)$ and $c' = c \circ e$ with $e \in E_K$ realized by the isometry ε. Then $V[c(t)] \mapsto V'[c(t)]$ is $g'\varepsilon^{-1}g^{-1}$, an isometry. Both facet sets lie in hyperplanes at the same distance $\sqrt{r^2 - r_K^2}$ from the center O, so it extends to a linear isometry A of $\mathbb{R}^4$. (2) Let f' share a ridge ρ (at least three non-collinear points) with f. Project to the 2-plane W orthogonal to the directions of ρ: ρ becomes a point p, and the hyperplanes through ρ become lines through p, tangent at their feet $T = dn$ to circles about O. The center O is interior to the vertex figure (Lemma 2.4), so it lies in the open wedge at p bounded by the two facet lines (angle below π). Each facet's circumcenter lies on its ray; it is relatively interior to the facet (the

facet is the vertex figure of a uniform 4-polytope), so $T \neq p$. Hence the two feet lie strictly on opposite sides of the line pO, and the hyperplane of f' is the tangent line from p at distance $d_{K_{f'}}$ on the side opposite to f's foot; it is the same for AV and V'. (3) Inside that hyperplane $AV[S_{f'}]$ and $V'[S_{f'}]$ are related by an index-preserving isometry, as in (1), which fixes ρ pointwise and the common circumcenter, which is not in the plane of ρ; so it is the identity. (4) The facet graph of V through ridges is connected, so $AV = V'$. □

Lemma 4.11 (Coset comparison)**.** *Let S be the full star at v_0 of a hypothetical convex uniform polytope of circumradius R, centered at the origin, and let T be a group of exact linear isometries fixing S. At a neighbor w, let $g_1, \dots, g_k$ be a complete, nonempty list of candidate linear isometries for its star, with a true congruence g_i among them. Suppose that for every j there is $\tau_j \in T$ and a certified Frobenius-norm bound $\|g_1\tau_j - g_j\|_F \leq \varepsilon_j$, and $2R\max_j \varepsilon_j < 1/2$. Suppose each g_j has anchor vertices a in S spanning $\mathbb{R}^5$, and every image $g_j(a)$ belongs to a known facet through w. Then every candidate gives the same full star at w.*

Proof. The identity

$$g_i\tau_i^{-1}\tau_j - g_j = (g_i - g_1\tau_i)\tau_i^{-1}\tau_j + (g_1\tau_j - g_j)$$

bounds the operator norm by $\varepsilon_i + \varepsilon_j$. On an anchor of norm R, the two images are less than $1/2$ apart. Both are vertices of the hypothetical polytope: one lies in the true star $g_i(S)$, the other in the known facet. By Lemma 2.2 they coincide. The spanning anchors determine the isometry, so $g_j = g_i\tau_i^{-1}\tau_j$ and $g_j(S) = g_i(S)$. □

4.4.3. *The one other vertex figure.* In the class whose scalar root is $R = 1$, with interval

$$[0.99999999999999999206,\ 1.00000000000000000867],$$

the class has two candidate vertex-figure signatures with the same facet multiset: two 5-cells (`A40001`), six $\{3\} \times \{3\}$ and eight tetrahedral prisms (`pr-tet`), 8 vertices. One is the vertex figure of the stericated 5-simplex `A510001`; the other is the only candidate signature in Case A not identified with a known polytope. No exact star with the latter encoding is asserted to exist; the refutation below applies to any hypothetical actual star it represents throughout the class interval.

Certified computation 4.12 (DEV: the extra vertex figure is refuted)**.** No convex uniform 5-polytope has an actual star represented by this extra candidate encoding in its class interval.

What is checked (independent checker, `develop.py`, Section 4.5.3). A *development*: starting from the star of v_0 (all facets with their full vertex sets), at the next vertex w without a star every congruence g with $g(v_0) = w$ mapping a facet of the star onto a known facet through w is tried (all of them: a star facet of the right type composed with every certified template symmetry taking the base to the right vertex); g is kept unless a known facet through w is certified not to lie in g(star) or adding g(star) gives a certified violation (separation, convexity, a cell in three facets, two facets with five affinely independent common vertices). Kept stars are compared by exact encodings, not by interval overlap; distinct stars give branches. Result: the signature has one retained rooted encoding. Any actual star represented by it, if one existed, would force its neighbors; it matches no listed polytope (`A510001`, with the same facet multiset, is excluded for all 48 frames; the other names of the class have 12 or 6 neighbors), the development fixes the star at the first neighbor, and at the second neighbor none of the 48 candidate congruences contains every facet already known there (a certified mismatch of a facet center or of a template map). So the vertex figure is `REFUTED`. *Uncertainty.* The development raises a distinct exception for an undecided identity, which contributes `OPEN` (also when another candidate survives) and is counted with its reason, so it can never yield `REFUTED`; candidate-point lookup tests every potentially matching point. The final guarded runs refute this vertex figure at $P_{\mathrm{ctl}} = 6$, 10 and 16 with zero search, development and model-guard uncertainty. *Second implementation:* `checks/independent-s23/` refutes the same signature at $P_{\mathrm{ctl}} = 6$, 10 and

16 by a finite development tree in which every branch dies (Section 4.5.4); its restricted $R = 1$ control (5-cell, $\{3\} \times \{3\}$ and tetrahedral prism only) identifies the other signature with `A510001` and refutes this one.

4.5. **Independent replays.** The implementations use the shared geometric specifications and inputs listed in Section 6.2. The stage-2 and stage-3 checkers below use `mpmath.iv`; the second implementation uses Arb. Agreement therefore checks distinct code and arithmetic paths, subject to those shared mathematical premises.

4.5.1. *Stage 1.* The stage-1 proof is replayed by `checks/independent-s1-b2/`, using exact constructions and certified closure bounds as described in Section 4.5.4.

4.5.2. *Stage 2:* `check_u2/`. The proof route of Computation 4.6. An orthogonal-frame gluing search with fail-first ridge choice in `mpmath.iv` at 200 bits. Each VF(K) is built from the guarded exact construction model (`exact_types.py`, Section 3.3); reductions are only the exact equivalences of Lemma 4.7, certified by `equivalences-p16.json` (Section 4.2). Branches are cut only for certified separation ($0 < |x - y|^2 < 1$), certified convexity violations or combinatorial contradictions; two points are identified only when $|x - y|^2 < 1/4$ is certified, and an enclosure that permits both equality and distinctness stops the class as undecided; a facet type is dropped when $R_K \geq R$ is certified. Every completed figure is rechecked (every ridge paired, every edge cycle allowed, every point not on a facet certified strictly inside: 0 exceptions). Result: 168, 332 and 629 signatures at $P_{\text{ctl}} = 6$, 10 and 16, with zero undecided classes (Section 4.2). Corruptions: deleting an allowed cycle removes exactly the vertex figure using it; substituting a facet type in a cycle adds no vertex figure; 905 equivalence-certificate corruptions (Section 4.2).

4.5.3. *Stage 3:* `check_u3/`. The stage-3 checker (Proposition 4.14 below) compares stars by exact encodings; the following lemma says what an encoding records and why equal encodings give equal stars.

Lemma 4.13 (Rooted determinism)**.** *Fix a common circumcenter c and an admissible radius R (a radius in the class interval with $R > R_K$ for every facet type K used). A* rooted encoding *of a star at v_0 records the full template type of an initial facet and its root frame (the images of the template's base vertex and its neighbors), and, along a connected traversal of the facet adjacency graph of the vertex figure (facets adjacent through cells containing v_0), the type of each further facet and the ordered correspondence of its model neighbor indices, taken modulo the exact stabilizer of the base vertex in the* full *template. If two stars at the same R have equal rooted encodings in root frames whose identification also identifies c, they are equal as sets of full facets.*

Proof. The root frame fixes the template map on the base vertex and its neighbors, which affinely span the facet hyperplane, so it fixes the full root facet. Suppose a facet F is fixed and $C \ni v_0$ is a cell of F at which the traversal reaches the next facet F' of type K'. The affine hull of C is fixed. The center of F' is at distance $\sqrt{R^2 - R_{K'}^2} > 0$ from the fixed common center c, so the hyperplane of F' is one of the two hyperplanes through aff C at that distance, and, as in step (2) of the proof of Lemma 4.10 one dimension up (the center of the polytope is interior, the facet centers are relatively interior by Lemma 2.1), it is the one whose foot lies on the side opposite to that of F. The ordered correspondence fixes the images of v_0 and of the vertex-figure points of C; together with the center of F', which is not in aff C, they affinely span the hyperplane of F', so the isometric template map, and hence the full facet F', is unique. Changing a correspondence by an element of the full-template base stabilizer does not change the placed full facet. The facet graph of the vertex figure is connected, so induction reaches every facet. If the stars are given as functions of R on an interval, the conclusion is equality of the placement functions at each common admissible R; the unique-root step then selects one value. □

Neither the coarse signature nor the larger group E_K of isometries of the vertex figure X_K would do here: for the snub-cube prism a mirror fixes the vertex figure but not the full facet (after reflecting one snub-cube-prism facet of $\{3\}\times$ snub cube, the two facets contain distinct vertices only 0.2956 apart). `checks/check_u3` (`canon.py`, `search.py`) uses full-template stabilizers and records the ordered model-neighbor indices, as the lemma requires.

(1) *Realization coverage.* For each signature, every actual star at a radius in the class interval is represented by a retained candidate encoding: facets are placed copies of templates through v_0 at the right distance; at an open cell every type with a cell of the same label is glued on the other side (convexity forces the side), under every rotation or reflection of the cell's vertex-figure polygon (reflections give mirror images of chiral facets); options equal under the stabilizer of the base are merged exactly; cuts only for certified reasons (separation, convexity, non-orthogonality, polygon or cell-patch mismatch, cell in three facets, two cells sharing three vertex-figure points, type counts, word multisets). A facet is rejected for a non-positive squared height only when the *upper* end of its enclosure is non-positive; an enclosure straddling zero is undecided. Every undecided exit, at the start or deeper in the search, is counted and makes the class `OPEN`; a class with an unresolved realization search cannot be used for identification by elimination. Point resolution tests every candidate enclosure by interval distance. Complete candidate states are grouped by an exact rooted encoding; actual stars with equal encodings in identified root frames are equal at every common admissible radius (Lemma 4.13). Result: one retained encoding per signature (629; 633 complete search states). Compatible interval constraints and the final closure/off-facet checks do not prove simultaneous exact satisfaction of all assembly equations. In particular, retaining a row is not an existence theorem.

(2) *Radius.* For one polygon, $f(R_{\mathrm{lo}})$ and $f(R_{\mathrm{hi}})$ have certified opposite signs and f' (forward-mode derivative on the whole class interval) excludes 0: a unique root in the class interval. The template angles and closure values use the proved inverse-cosine enclosure, and the class intervals are the exported stage-1 classes, with the coverage audit of Computation 4.9. This certifies the scalar root, not existence of an assembled star at that root.

(3) *Identification by elimination.* A listed name Q enters a class only if its radius lies in the class interval and every facet type of Q, derived from its name by construction rules, is a facet type of the input; otherwise Q is only reported. For each retained encoding e of the class, possible matches to Q are tested: `NO` if for every frame (isometry fixing the center with $v_0 \mapsto q_0$, fixed by v_0 and four vertex-figure points sent to every neighbor 4-tuple of q_0 with the same distances) some star point is certified apart from every vertex of Q. A frame is skipped only when an exact interval Gram entry is disjoint from the required one (the double-precision Gram comparison only chooses which entry to test); a star point is certified apart only by coordinate-interval disjointness from every local vertex of Q or a positive interval lower bound on its squared distance; local vertices of Q are discarded only when an interval lower bound exceeds the cutoff; a frame that cannot be inverted or decided is `UNDECIDED`, never `NO`. `NO-COUNT` if e has a number of vertex-figure points different from the exact neighbor count of q_0 (an upper bound from enclosures containing 1, a matching lower bound exhibited by the exact model of Q's construction). An actual star represented by e is identified with Q when every other encoding of the class is excluded for Q: the separately constructed Q has an actual star represented in this class by completeness of stage 2 and step 1, so its encoding must be e. Step 2 fixes the common radius and Lemma 4.13 gives equality in the identified root frames. No interval overlap is used as an equality; the overlap test `MATCH` is reported as a consistency check only.

(4) *Local uniqueness.* At each neighbor w, every congruence $g = [L_{F^*}S \mid \cdot][L_F \mid \cdot]^T$ (F a star facet of the type of a facet F^* through v_0 and w, S a template symmetry taking the base to w's template vertex, all of them) not certified to miss a facet through w is kept; all kept g must give one exact encoding, hence at most one actual full star at w at the fixed radius

(Lemma 4.13); an empty kept set is not accepted as forcing. For an actual polytope the star at w supplies existence. Whether the facets of g(star) can match the known facets through w is decided by a perfect matching in the complete bipartite graph of interval-compatible pairs (`locuniq.py`); a candidate is excluded only if no perfect matching exists.

(5) *Development* for any actual star represented by the unidentified encoding, as in Computation 4.12.

Template facts (neighbors, cells, symmetries of the 181 facet types) are certified against exact models of the constructions: symmetries by construction, vertices at exact distance 1, exactly flat cell vertex sets, `sadi` and `gap` in $\mathbb{Q}(\sqrt{5})$; all symmetry candidates are enumerated and rejected only for certified reasons, and the accepted sets equal the exact groups (orders $|\mathrm{Stab}| \cdot |V|$: 120 or 240 for A_4 types, 384 or 1152 for B_4, 1152 or 2304 for F_4, 14400 for H_4, $4pq$ or $8p^2$ for duoprisms, $8n$ for antiprism prisms, 48, 96, 120 or 240 for polyhedral prisms, 400 for `gap`, 576 for `sadi`). Double precision is used only to propose, except in one place covered by a stated error lemma with asserted preconditions (a coordinate lemma: a double within $1.1 \cdot 10^{-6}$ of an exact coordinate of size at most 64, used to certify a rejected symmetry candidate's image apart from distant template vertices). Frames and local vertices are excluded by interval evidence only. All models are built and used under the runtime guard of Section 3.3. Result at $P_{\max} = 16$: 629 retained encodings, 628 `KNOWN` (628 distinct constructed polytopes, all 890 names matched, none outside the library), 1 `REFUTED` (the extra candidate), 0 `OPEN`; scalar-root uniqueness and conditional forcing for all 629; 230 `NO-COUNT` and 32 `NO` eliminations. Corruptions: shifting every class interval by 10^{-12} leaves no retained encoding and no `KNOWN`; mirroring one snub-cube-prism facet changes `KNOWN` to `REFUTED`. *Runs with the model guard active*: $P_{\mathrm{ctl}} = 6$: 136 classes, 168 encodings, 167 `KNOWN`, one `REFUTED`; $P_{\max} = 10$: 278, 332, 331, one; $P_{\max} = 16$: 563, 629, 628, one; all roots and all forcing checks certified, and zero `OPEN`, search, development and model-guard uncertainty. The checker's inputs are the completeness of stage 2 (Computation 4.6), the construction properties of Section 3.3, the separation lemma, and the standard facts used by its certificates.

Proposition 4.14 (stage 3, proof route)**.** *Let P be a convex uniform 5-polytope with all 2-faces at most 16-gonal, and let σ be the signature of its vertex figure at v_0, in the surviving class I containing R. If σ has verdict `KNOWN` with name Q, then $P \cong Q$; if σ has verdict `REFUTED`, P does not exist.*

Proof. Let X be the actual star of P at v_0. By step 1 it is represented by the single retained encoding of σ; by step 2, R is the unique root on the whole class interval. If the verdict is `KNOWN`, then Q's own star is a realization of a signature of the class (stage 2 and step 1 applied to Q, whose facets lie in the library) and every other encoding of the class is excluded for Q (step 3; Q entered the class because its radius lies in I). Its star therefore has the same encoding as X and, by step 2, the same radius; Lemma 4.13 makes the stars equal in identified root frames. Thus after an isometry P and Q have the same star X at v_0. By step 4 any convex uniform 5-polytope with star X at v_0 has one and the same star at every neighbor of v_0; Lemma 2.11 gives $P = Q$ after the isometry. If the verdict is `REFUTED`, the development of step 5 excludes every possibility (Computation 4.12). □

4.5.4. *Independently written second implementations.* The packages `checks/independent-s1-b2/` and `checks/independent-s23/` implement separate algorithms using Arb (python-flint) and SymPy. The latter shares stage-1 classes, model certificates, known-name lists and the facet-name table with the other stage-2/3 implementation; neither package uses `mpmath.iv`.

Stage 1 and B2: `checks/independent-s1-b2/`. Library: exact parabolic Coxeter combinatorics with rational coefficient-pair keys, exact supporting planes for the two 600-cell deletions, product identities; 181 types, 558 words, 1,277 occurrences, all enclosures inside the recorded ones. Stage 1: closed walks of length three to six, exact Gram matrices for three rays, the two-endpoint monotonicity certificate of Section 4.1, exact phase products for flatness; the same 14,965 cycles and

statuses, 7,080 flat, 57 degenerate, 537 without roots, 7,291 roots each inside the recorded interval. B2: Section 5, Computation 5.8. The derivative of the closure-angle cosine is checked symbolically and its strict sign certified at both ends of the admissible interval for all 558 words; a synthetic word with a sign change is rejected.

Stages 2 and 3: `checks/independent-s23/`. Stage 2 as described in Section 4.2. Stage 3: closure constants derived from the full exact models; radius uniqueness on the whole exported class interval by Arb inverse cosine and forward-mode derivatives in $s = 1/R$; full five-dimensional stars rebuilt from the base and four model neighbors, with every facet expanded by all exact right cosets of the full-template base stabilizer in the isometry group of its vertex figure (restoring chiral orientation alternatives); candidate states grouped only by correspondence isomorphisms whose facet maps are certified full-template stabilizers; identification by exhaustive elimination (certified Gram disjointness or strict apartness from every vertex of Q, exact neighbor counts); local uniqueness by all congruences mapping a star facet to a known facet through w, complete bipartite matching and the coset bound of Lemma 4.11 with Arb norms; development by a finite tree in which every branch must die; a grid index on exact dyadic endpoints that falls back to all points rather than excluding. Undecided identities, signs or resource limits make a class `OPEN`. *Result*: at $P_{\mathrm{ctl}} = 6$, 10 and 16 all 136/278/563 classes complete with 168/332/629 signatures; stage 3 finds 167/331/628 signatures `KNOWN` (235/455/890 names matched) and one wholly refuted signature at each bound (the extra vertex figure), with no `OPEN` class or realization. Its raw candidate-state counts are 176/348/657 against `checks/check_u3`'s 168/332/629: the restored mirror placements (8/16/28) each have a strict full-star separation contradiction; all classification conclusions are the same. *Shared with* `check_u2/checks/check_u3`: the stage-1 class intervals (regenerated as black boxes and matched to the 30 input pins), the known-name lists and facet-name table, the construction theorem (`checks/model_audit/wythoff_proof.md`), the two exceptional model certificate files, the alias table and the separation lemma.

4.6. **Proof of Theorem 1.2.** Let P be a convex uniform 5-polytope with all 2-faces at most 16-gonal, scaled to unit edges, with circumradius R and a vertex v_0. By Lemma 4.4 all cycles at v_0 lie in the surviving class I containing R (Computations 4.1, on the input certified by Computation 3.5, and 4.5). Its vertex figure satisfies (a)–(d) of Section 4.2, so its signature is one of the signatures listed for I (Computation 4.6, proof route `check_u2` with the exact equivalence certificate). Each listed signature has verdict `KNOWN` with a name $Q \in \mathcal{K}_{16}$ or is the refuted extra vertex figure, so Proposition 4.14 gives $P \cong Q$ or a contradiction. Each of these computations has a second rigorous implementation that reaches the same result, sharing the inputs listed in the shared-input discussion in Section 6.2 (see also Section 4.5.4). □

TABLE 1. Counts of the finite computation for the control libraries $\mathcal{H}_6$, $\mathcal{H}_{10}$ and $\mathcal{H}_{16}$ as checked by the full replays. Retained signatures are not exact-existence certificates for every row.

	$\mathcal{H}_6$	$\mathcal{H}_{10} = \mathcal{L}_{10}$	$\mathcal{H}_{16} = \mathcal{L}_{16}$
canonical facet types	76	106	181
cycles	2,559	5,471	14,965
flat (exact) / degenerate roots (exact)	790 / 23	1,703 / 39	7,080 / 57
radius classes / surviving the filter	760 / 136	1,790 / 278	4,224 / 563
surviving classes with a known radius	135	277	562
vertex-figure signatures	168	332	629
distinct known polytopes	167	331	628
branches of the extra vertex figure (stuck)	4	4	4

4.7. **Computed coverage.** The guarded searches at $P_{\text{ctl}} = 6, 10, 16$ give 136/278/563 surviving classes and 168/332/629 strict vertex figures, with zero undecided or non-strict results. The independent stage-1/B2 replay and stage-2/3 replays of Section 4.5.4 agree. The full $P = 16$ computation covers all 563 surviving classes; its 629 candidate figures lead to 628 known polytopes and the refuted extra candidate. These are coverage statements, not assertions that every retained local encoding has an exact realization.

At $P_{\text{ctl}} = 6$ ten known radii lie in no surviving class and four known facet multisets are not found; all of them have octagons or decagons, in facets outside $\mathcal{H}_6$ (the variable families remain capped), so these lists are non-empty by design; at $P_{\max} \geq 10$ both are empty.

5. Case B: a 2-face with at least 17 sides

5.1. **Setting.** In this section P is a convex uniform 5-polytope with unit edges, circumradius R, center c, having a large 2-face (at least 17 sides). By vertex-transitivity every vertex lies in a large 2-face. Fix a vertex v; VF is its vertex figure, a convex 4-polytope inscribed in a 3-sphere of radius $r = \sqrt{1 - 1/(4R^2)} < 1$ (Lemma 2.4). Faces of VF correspond to faces of P at v: VF-vertex = edge of P at v (we identify it with the neighbor), VF-edge = 2-face, VF-ridge = cell, VF-facet = facet. A VF-edge is *long* if its 2-face is large; its length is $\text{ch}_n \geq \text{ch}_{17} = 2\cos(\pi/17) = 1.96595\ldots$. A facet is *large* if it contains a large 2-face. For a large 2-face $F \ni v$ with VF-edge e, the *fan* of F (or of e) is the set of VF-facets $\text{VF}(K)$ with $K \supset F$, and its vertices are the *fan vertices*.

5.2. **Facts about the families.**

Fact 5.1 (F2)**.** *Let $q \notin \{3, 4, 5, 6, 8, 10\}$. A convex uniform polyhedron with a q-gonal face is P_q or A_q; a convex uniform 4-polytope with a q-gonal 2-face is a duoprism $\{q\} \times \{p\}$ ($p \geq 3$, including $P_q \times \{\} = \{q\} \times \{4\}$) or $A_q \times \{\}$ (Facts 1.4 and 1.5). In particular this holds for every large q, and for $q \in \{7, 9, 11, \ldots, 16\}$ ("family polygons"), which (P1) uses.*

Fact 5.2 (F3)**.** *In $\{n\} \times \{q\}$ and in $A_n \times \{\}$ every vertex lies on an n-gon. At a vertex v: $\text{VF}(\{n\} \times \{q\})$ is the tetrahedron $\text{conv}(a, b, g, g')$, where a, b are the neighbors on the n-gon and g, g' those on the q-gon: $|ab| = \text{ch}_n$, $|gg'| = \text{ch}_q$, $ab \perp gg'$, and the other four edges have length $\sqrt{2}$. $\text{VF}(A_n \times \{\})$ is the pyramid with apex z (the prism edge) over the trapezoid $(a, b, y, x) = \text{VF}(A_n)$: ab is the VF-edge of the n-gon, $|by| = |yx| = |xa| = 1$ (the three triangles at v), the diagonals ay, bx have length $\sqrt{1 + \text{ch}_n}$, and the four apex edges have length $\sqrt{2}$.*

These follow directly from the stated product and antiprism constructions.

Lemma 5.3 (F5: cycle types)**.** *Let F be a 2-face of P with $n \notin \{3, 4, 5, 6, 8, 10\}$ sides. Then the cycle of F is one of*

(i) $\{n\} \times \{q_1\}, \ldots, \{n\} \times \{q_k\}$ *(all cells P_n), with $k \in \{3, 4, 5\}$ and $\sum_i (1 - 2/q_i) < 2$;*
(ii) $A_n \times \{\}$, $A_n \times \{\}$, $\{n\} \times \{q\}$;
(iii) $A_n \times \{\}$, $A_n \times \{\}$, $\{n\} \times \{q_1\}$, $\{n\} \times \{q_2\}$ *with $\{q_1, q_2\} \in \{\{3, 3\}, \{3, 4\}, \{3, 5\}\}$.*

Proof. By Fact 5.1 the cells at F are P_n or A_n and the facets $\{n\} \times \{q\}$ or $A_n \times \{\}$. The face angles of the tangent cone at F (Lemma 2.6) are $\pi(1 - 2/q)$ for $\{n\} \times \{q\}$ (between two P_n) and $\pi/2$ for $A_n \times \{\}$ (between A_n and P_n), with sum below 2π. Both neighbors of an A_n ray are $A_n \times \{\}$, each containing one A_n cell at F, so j rays of type A_n cost $2j$ distinct right angles, and $j \leq 1$. If $j = 0$ we get (i), with $k \geq 3$ and $k \leq 5$ since each term is at least $1/3$. If $j = 1$ there are $m \geq 0$ duoprisms between the two P_n rays, and the cycle has $m + 2$ rays. The tangent cone at F is a full-dimensional pointed cone in the 3-dimensional space N, so a section of it by a positive affine functional is a convex polygon and the cone has at least three extreme rays; hence $m = 0$ is impossible. $m = 1$ is (ii); $m = 2$ needs $1/q_1 + 1/q_2 > 1/2$, i.e. (iii); $m \geq 3$ gives a sum of at least $\pi + 3\pi/3$, too large. □

Lemma 5.4 (Long edges are disjoint)**.** *Two long VF-edges share no vertex.*

Proof. A long edge $[a,b]$ subtends at the center of the VF sphere an angle θ with $\sin(\theta/2) = \cos(\pi/n)/r \geq \cos(\pi/17)$, so $\theta \geq \pi - 2\pi/17$ (158.82°). If $[a,b]$ and $[a,b']$ are both long, b and b' lie within $2\pi/17$ (21.18°) of the antipode of a, so $|b-b'| \leq 2r\sin(2\pi/17) < 0.7225 < 1$, contradicting Lemma 2.2. □

Lemma 5.4 is used in Section 5.6.2: the Arb certificate of (W) leaves out the ends of the long edges themselves, and this lemma covers them.

5.3. **The radius forced by a large 2-face.** Let F be an n-gon of P at v, $n \notin \{3,4,5,6,8,10\}$ (so that Lemma 5.3 applies), with center o_F in the plane L, and write $c = o_F + W$ with $W \perp L$; so $R^2 = r_n^2 + |W|^2$. A neighbor $v+u$ of v with $u \perp L$ is on the circumsphere iff $u \cdot W = 1/2$ (expand $|v+u-c|^2 = R^2$ using $u \cdot (v - o_F) = 0$).
Type (i). The cells at F are $F \times [v, v+u_i]$ with unit $u_i \perp L$; consecutive u_i, u_{i+1} span the q_i-gon of $\{n\} \times \{q_i\}$ at v, so $|u_i - u_{i+1}| = \mathrm{ch}_{q_i}$. All u_i satisfy $u_i \cdot W = 1/2$, so they lie on a circle of radius $S = \sin\alpha$ on the cone of half-angle α about W, $\cos\alpha = 1/(2|W|)$. By Lemma 2.6(c), W is strictly inside the cone spanned by the u_i, i.e. the circle's center is strictly inside the inscribed polygon of the u_i: all central angles $2\arcsin(\cos(\pi/q_i)/S)$ are below π and they sum to 2π. Let $f_c(S) = \sum_i 2\arcsin(\cos(\pi/q_i)/S)$ on $[c_{\max}, 1]$, $c_{\max} = \max_i \cos(\pi/q_i)$; f_c is strictly decreasing and $f_c(1) = \sum \pi(1-2/q_i) < 2\pi$. Hence:

Lemma 5.5 (Type (i) radius and closability)**.** *A configuration $c = (q_1, \ldots, q_k)$ occurs in a type-(i) cycle only if it is* closable*: $f_c(c_{\max}) > 2\pi$. Then $f_c(S) = 2\pi$ has a unique solution $S \in (c_{\max}, 1)$, the* configuration radius *$R_c = 1/(2\sqrt{1-S^2})$ depends only on the multiset c, and*

$$R^2 = r_n^2 + R_c^2.$$

*Moreover $R_c \geq r_{q_{\max}}$. For $k=3$, closability is equivalent to the triangle with sides $\mathrm{ch}_{q_1}, \mathrm{ch}_{q_2}, \mathrm{ch}_{q_3}$ being acute; for $k=4$ to the circle's center being strictly inside the inscribed quadrilateral. If $f_c(c_{\max}) = 2\pi$ exactly (*degenerate*), the center lies on the hyperplane of a facet, which is impossible.*

Proof. All but the formula $R_c \geq r_{q_{\max}}$ and the triangle criterion is above. $S \geq \cos(\pi/q_{\max})$ gives $R_c^2 = 1/(4(1-S^2)) \geq 1/(4\sin^2(\pi/q_{\max}))$. For three rays: the foot of the perpendicular from 0 to the plane of $u_1u_2u_3$ is the circumcenter o_T of the chord triangle, and $o_T \cdot u_i = |o_T|^2$ for each i, so $W = o_T/(2|o_T|^2)$ (both satisfy the three equations $x \cdot u_i = 1/2$, which have a unique solution); so W is interior to the cone iff o_T is interior to the triangle iff the triangle is acute, and a right triangle puts W in the plane of a facet. The same argument with the circle gives the $k=4$ form. □

Remark 5.6 (four or five rays). For $k \geq 4$ a configuration is never degenerate, and it is closable if and only if $\sum_i (1-2/q_i) < 2$. Indeed, at $S = c_{\max}$ one maximal term of f_c equals π, and each of the other $k-1 \geq 3$ terms is strictly larger than $2\arcsin(\cos(\pi/q_i)) = \pi(1-2/q_i) \geq \pi/3$ because $c_{\max} < 1$; so $f_c(c_{\max}) > 2\pi$ even when maximal entries are tied. Since f_c is strictly decreasing, a root in $(c_{\max}, 1)$ exists exactly when $f_c(1) = \pi\sum(1-2/q_i) < 2\pi$ (at equality the only root is $S=1$, which gives no finite R_c). This makes the closability classification used in B2 and (P1) checkable by hand for $k \geq 4$. The exact angle condition already excludes $k \geq 6$, so the finite enumerations need $k = 3,4,5$ only; the B2 checker rejects cycles with fewer than three or more than five entries, and the (P1) closure routines reject fewer than three, explicitly.

Exactly three configurations with entries at most 16 are degenerate: $(3,3,4)$, $(3,4,6)$, $(3,5,10)$, since $\mathrm{ch}_3^2 + \mathrm{ch}_3^2 = \mathrm{ch}_4^2$, $\mathrm{ch}_3^2 + \mathrm{ch}_4^2 = \mathrm{ch}_6^2$ and $\mathrm{ch}_3^2 + \mathrm{ch}_5^2 = \mathrm{ch}_{10}^2$ (the last is $1+(3+\sqrt{5})/2 = (5+\sqrt{5})/2$). The configuration $(3,5,q)$ is not closable for $q \geq 11$, since $1+\mathrm{ch}_5^2 = 3.618\ldots < \mathrm{ch}_q^2$; similarly $(3,3,q)$ for $q \geq 5$ and $(3,4,q)$ for $q \geq 7$. For c'' a sequence such that (n, c'') is closable, write

$$\delta_{c''}(n) := R^2_{(n,c'')} - r_n^2.$$

Lemma 5.7 (closed forms)**.** *Let $n \geq 17$, $\theta = \pi/n$, $c = \cos\theta$ and $x = c^2$. Then*

$$\delta_{(4,4)}(n) = \tfrac{1}{4}, \qquad \delta_{(3,6)}(n) = \frac{x}{4x-1} = \frac{1}{4-\sec^2\theta}, \qquad \delta_{(3,3,3)}(n) = \frac{2c+1}{8(1+c)} = \frac{h_A(n)^2}{4}.$$

$\delta_{(3,6)}$ decreases and $\delta_{(3,3,3)}$ increases with n, so for $n \geq 17$, including the limit $n = \infty$, $\delta_{(3,6)}(n) \in [1/3,\ \delta_{(3,6)}(17)]$ with $\delta_{(3,6)}(17) = \cos^2(\pi/17)/(4\cos^2(\pi/17) - 1) = 0.3372617\ldots < 0.337262$, and $\delta_{(3,3,3)}(n) \in [h_A(17)^2/4,\ 3/16]$ with $h_A(17)^2/4 = 0.1869633\ldots$; the value $3/16$ is attained only in the limit $n = \infty$.

Proof. For a chord triangle with squared sides A, B, C the circumradius satisfies $S^2 = ABC/((A+B+C)^2 - 2(A^2+B^2+C^2))$, and $R_c^2 = 1/(4(1-S^2))$, $r_n^2 = 1/(4(1-x))$. For $(n,3,6)$ the sides are $2c, 1, \sqrt{3}$, which gives $S^2 = 3x/(-1+8x-4x^2)$ and $\delta_{(3,6)} = x/(4x-1)$; the triangle is acute since $1/2 < x < 1$, so this is the closable solution. For $(n,4,4)$ the sides are $2c, \sqrt{2}, \sqrt{2}$, giving $S^2 = 1/(2-x)$ and $\delta = 1/4$ (the prism P_n); this triangle is also acute. For $(n,3,3,3)$ put $S^2 = 1/(3-2c)$ and $\sin\alpha = 1/(2S)$; then $\pi/6 < \alpha < \pi/4$, the three unit chords subtend 2α each, the remaining arc $2\beta = 2\pi - 6\alpha$ lies in $(\pi/2, \pi)$, and $\sin\beta = \sin 3\alpha = \sin\alpha(3 - 4\sin^2\alpha) = 2c\sin\alpha$, so its chord is $2S\sin\beta = 2c$: all four arcs are below π and sum to 2π. Substitution gives $\delta_{(3,3,3)} = (2c+1)/(8(1+c)) = \big(1-1/(2(1+c))\big)/4 = h_A(n)^2/4$ (the antiprism A_n), using $\cos^2(\theta/2) = (1+c)/2$. Finally $d\delta_{(3,6)}/dc = -2c/(4c^2-1)^2 < 0$ and $d\delta_{(3,3,3)}/dc = 1/(8(1+c)^2) > 0$, and c increases to 1 with n. □

The limiting value $3/16$ is included in the closed range used by (P1). The partner filter uses the valid bound 0.3373 for $\delta_{(3,6)}(17)$; the exact rational function, rather than a rounded constant, is evaluated throughout the parameter interval.

Type (ii). The A_n cell has its second n-gon at height $h_A = h_A(n)$ along a unit vector $\hat{h} \perp L$, rotated by π/n. The edge condition gives $h_A^2 = 1 - 1/(4\cos^2(\pi/(2n)))$, and equidistance from the two n-gons gives $W \cdot \hat{h} = h_A/2$, so $R_{A_n}^2 = r_n^2 + h_A^2/4$. The other two rays z_1, z_2 (lateral edges of the P_n cells) make right angles with $\hat{h}$ and the angle $\pi(1-2/q)$ with each other, so they lie in the plane orthogonal to $\hat{h}$ in $L^\perp$; the component W' of W there satisfies $W' \cdot z_j = 1/2$ and is the center offset of the q-gon, $|W'| = r_q$. Hence

$$R^2 = R_{A_n}^2 + r_q^2.$$

Type (iii). The rays are $\hat{h}, z_1, w, z_2$. As in (ii), $W = (h_A/2)\hat{h} + W'$ with $W' \in \operatorname{span}(z_1, z_2)$, $W' \cdot z_j = 1/2$, so with ϕ the angle between z_1 and z_2 and $\gamma := \cos\phi$, $|W'|^2 = 1/(2(1+\gamma))$ and $R^2 = R_{A_n}^2 + 1/(2(1+\gamma))$. Writing $w = \alpha z_1 + \beta z_2 + \eta\hat{h}$ with $w \cdot z_j = c_j := \cos(\pi(1-2/q_j))$, the remaining condition $W \cdot w = 1/2$ reads $h_A\eta = 1 - (\alpha+\beta)$, where $\alpha + \beta = (c_1+c_2)/(1+\gamma)$ and $\eta^2 = 1 - |\alpha z_1 + \beta z_2|^2$. Convexity of the cone $(\hat{h}, z_1, w, z_2)$ puts w and $\hat{h}$ on opposite sides of $\operatorname{span}(z_1, z_2)$, so $\eta < 0$ (the *geometric* branch). Squaring, with $N = 1 - \gamma^2 - c_1^2 - c_2^2 + 2\gamma c_1 c_2$ (so that $\eta^2 = N/(1-\gamma^2)$) and $M = 1 + \gamma - c_1 - c_2$, gives the polynomial equation

$$h_A^2\, N\,(1+\gamma) - M^2\,(1-\gamma) = 0,$$

to be solved on the domain $-1 < \gamma < 1$ together with the norm condition $N \geq 0$ and the unsquared sign $\eta < 0$. When $c_1 = c_2$ (the pair $(3,3)$) the polynomial has the spurious factor $1-\gamma$, which is canceled, leaving a quadratic; otherwise it is a cubic. The radius identity above holds at every geometric root. At $n = 17$, for instance, the roots in $(-1,1)$ are one for $(3,3)$ and two each for $(3,4)$ and $(3,5)$, and exactly one of them is geometric in each case. B2 checks *all roots in* $(-1,1)$ (Computation 5.8), so no claim about their number is needed. The (W) checker parametrizes the type-(iii) fan by $\cos(\phi/2)$ and recovers n from it, without root finding.

5.4. **Step B2: no large-polygon radius is a Case-A radius.**

Certified computation 5.8 (B2)**.** Let $n \geq 17$. None of the following radii lies in an interval of Ω (Definition 4.2):

- $\sqrt{r_n^2 + R_c^2}$ for every closable configuration c (any entries ≥ 3, $k \in \{3,4,5\}$, $\sum(1 - 2/q_i) < 2$);
- $\sqrt{R_{A_n}^2 + r_q^2}$ for every $q \geq 3$;
- $\sqrt{R_{A_n}^2 + 1/(2(1+\gamma))}$ for every root $\gamma \in (-1,1)$ of the type-(iii) polynomial equations, geometric or not.

Reduction to a finite check. r_q increases with q and $r_{398} > 63.344 > \max\Omega$. Every radius above exceeds r_n, so only $17 \leq n \leq 397$ matter (381 values). For type (i), $R_c \geq r_{q_{\max}}$ (Lemma 5.5), so configurations with an entry ≥ 398 give $R > \max\Omega$; for type (ii), $R \geq r_q$, so $q \leq 397$; for type (iii), $q_1, q_2 \leq 5$. Both `b2check.py` (`QMAX=397`, function `run`) and `checks/independent-s1-b2/b2.py` (function `run`) certify the r_{398} inequality before the enumeration. The boundary certificate `checks/independent-s1-b2/evidence/b2-comparison.json`, `cutoff`, records `last_finite_n=397`, `first_tail_n=398` and $r_{397} < \max\Omega < r_{398}$.

Proof route (`b2check.py`). This uses `mpmath.iv` at 200 bits, with its own derivation of the three radius formulas (as above). Type (i): all 1,304 closable multisets with entries 3..397, $k = 3,4,5$ and exact rational angle sum below 2 (1,316 cyclic arrangements; R_c depends only on the multiset), R_c by a verified bracket, for each of the 381 values of n (496,824 radii); closability by interval arcsine and the exact degenerate cases in $\mathbb{Q}(\sqrt{5})$. Type (ii): $q = 3, \ldots, 397$ (150,495 radii). Type (iii): every root in $(-1,1)$, isolated from certified nonzero endpoint values by subdivision; a box is discarded only for a nonzero interval polynomial value or a derivative of constant sign with equal strict endpoint signs, opposite strict signs with a derivative excluding zero certify one root, every split point is certified non-root, and an unresolved box is a failure. There is no positive-root or geometric-branch prefilter: 1,905 roots (1,143 geometric, 762 non-geometric), all compared with Ω. The overlap query is complete (sorted lower endpoints with a prefix maximum of upper endpoints). Result: 649,224 radii, **0 coincidences, 0 undecided boxes**. The closest approach is a relative gap of $1.66 \cdot 10^{-7}$ (type (i), $n = 24$, $c = (5,5,8)$, $R = 4.8302\ldots$), against interval widths of about 10^{-45} (radii) and 10^{-20} (Ω). The closability classification: not closable exactly $(3,3,q)$, $q \geq 5$; $(3,4,q)$, $q \geq 7$; $(3,5,q)$, $q \geq 11$; degenerate exactly $(3,3,4)$, $(3,4,6)$, $(3,5,10)$; all others closable (Remark 5.6). *Root coverage audit* (`check_root_audit.py`, importing neither enumeration program): it reconstructs each quadratic or cubic in 320-bit Arb arithmetic, counts its roots in $(-1,1)$ by Sturm sequences with all signs certified and no repeated roots, and checks that the certified brackets are disjoint, have opposite endpoint signs, the right derivative sign and branch label, and are as many as the Sturm count; so no root in $(-1,1)$ is omitted. It rejects four corruptions (a dropped negative non-geometric root, a duplicate, a changed branch, an endpoint bracket). The B2 check also returns zero coincidences against the independent stage-1 intervals.

Inverse sine. The lower-bound proposal loop of `rasin` uses an explicit guard: for a positive proposal require $0 \leq t_{\mathrm{lo}} \leq L$, where L is the lower endpoint of the interval for $\pi/2$. Hence t_{lo} is in the principal domain before sine monotonicity is used; a nonpositive proposal is replaced by the safe lower bound zero, and an unproved domain raises an error. The wrong-branch proposal $t_{\mathrm{lo}} = 4$ at $x = 1/2$ is rejected.

Second implementation. Written from this paper only and run against the independent Ω of that package (its own stage 1): all 381 values $17 \leq n \leq 397$, all 1,304 closable multisets (1,171 not closable and 3 degenerate, classified exactly), $q = 3, \ldots, 397$, and every root in $(-1,1)$ of the three type-(iii) polynomials found by exhaustive subdivision and counted by a separate Sturm sequence (1,143 geometric, 762 non-geometric); the cutoff $r_{397} < \max\Omega < r_{398}$ and the parameter tails are certified. Result: 649,224 radii, 0 coincidences, 0 undecided; the smallest absolute separation is $7.9977\ldots \cdot 10^{-7}$, again at $n = 24$, $c = (5,5,8)$. Positive controls 1,750/1,750, 196/196,

42/42 geometric and 2/28 non-geometric; 16 semantic corruptions behave as expected. *Shared with* `check_b2`*:* the reductions of this section only (the two use different Ω files, built by different stage-1 implementations).
Positive control. Every type-(i), (ii) and geometric type-(iii) radius with $n \leq 16$ and entries at most 16 is a stage-1 root, so it must lie in Ω: 1,750/1,750, 196/196 and 42/42 of them are found in Ω. In addition 2 of the 28 non-geometric type-(iii) roots in that range also lie in Ω (44/70 type-(iii) roots in total); these are not required. These are non-proof checks of the radius formulas. *Corruptions:* three intervals inserted into Ω around computed radii give exactly three coincidences; the slip $r_n = 1/(2\sin(2\pi/n))$ gives 1,000 coincidences (the radii of $\{n/2\} \times Q$). *Input identity:* Ω is exported from the unfiltered stage-1 output by `export_omega.py` with the degenerate-cycle guard described after Definition 4.2; the checker refuses an Ω whose SHA-256 differs from `c7773fab…89bc33` and also checks its content; all 7,080 infinite upper endpoints belong to flat cycles, so no surviving interval is unbounded (an unbounded interval would make B2 vacuous for large n).

Corollary 5.9. *Every 2-face G of P lies in a large facet. Consequently every VF-edge at v lies in a VF-facet* $\mathrm{VF}(K)$ *with K large, and K contains a large 2-face through v.*

Proof. If G is large there is nothing to prove. Otherwise, suppose no facet around G is large; then all of them have only polygons of at most 16 sides, so they are in $\mathcal{L}_{16}$ (Lemma 3.1) and $R \in \Omega$ by Corollary 4.3. But R is also the radius forced by the cycle of a large 2-face of P, which is of type (i), (ii) or (iii) (Lemma 5.3) with the parameters of Computation 5.8 (type (i) configurations are closable by Lemma 5.5; in type (iii), $\gamma = \cos\phi$ is a root of the polynomial in $(-1, 1)$, and B2 covers every such root), so $R \notin \Omega$. For the second part, a large facet K is $\{m\} \times \{q\}$ or $A_m \times \{\}$ with $m \geq 17$ (Fact 5.1), and every vertex of K, in particular v, lies on an m-gon of K (Fact 5.2). □

Let $s \geq 1$ be the number of large 2-faces at a vertex (the same at every vertex).

5.5. Exactly one large 2-face at each vertex.

Theorem 5.10. *If $s = 1$, then $P \cong \{n\} \times Q$ for a convex uniform polyhedron Q, or $P \cong A_n \times \{q\}$.*

Fix v, its large 2-face F (an n-gon with plane direction L) and $e = [a, b]$ its VF-edge. By Corollary 5.9 and $s = 1$, every VF-edge lies in a VF-facet $\mathrm{VF}(K)$ with $K \supset F$; so *every VF-edge, and hence every VF-vertex, lies in a facet of the fan of e.* The fan is given by the cycle type of F (Lemma 5.3).

Type (i). The fan facets are $T_i = \mathrm{conv}(a, b, x_i, x_{i+1})$, where $[v, x_i]$ is the lateral edge of the i-th cell $F \times [v, x_i]$; so $x_i - v \perp L$, and the neighbors of v are exactly $a, b, x_1, \ldots, x_k$. The x_i are distinct (two P_n cells at F with the same lateral edge span the same 3-space).
Product lemma. For a vertex t let F_t be its unique large 2-face. By vertex-transitivity, the local picture at t is the one at v (a symmetry mapping v to t maps F to F_t, by uniqueness). For a neighbor t' of t, either $t' \in F_t$, and then $F_{t'} = F_t$; or $t' - t$ is a lateral edge, orthogonal to the plane of F_t, and the duoprism $\{n\} \times \{q_i\}$ containing the edge tt' contains the translate $F_t + (t' - t)$, a large 2-face at t', so $F_{t'} = F_t + (t' - t)$. These translates come from the actual prism cells at the edge, not from the shape of the vertex figure. The edge graph of P is connected [5], so every F_t is a translate $F + \tau_t$ with $\tau_t \perp L$ (each step adds 0 or a vector orthogonal to L). Hence the vertex set is $F_0 + X$, with F_0 the polygon F centered at the origin of L and X a finite set in an affine copy of $L^\perp$, and $P = \mathrm{conv}\, F_0 + \mathrm{conv}\, X \cong \{n\} \times Q$ with $Q = \mathrm{conv}\, X$, a 3-polytope since $\dim P = 5$. Every point of X is a vertex of Q (a point (f, x) of a product is a vertex iff both factors are vertices). Q has unit edges and regular faces, because its edges and 2-faces G give edges and 2-faces $\{f\} \times G$ of P; a symmetry g of P maps F_t to $F_{g(t)}$, so its linear part preserves L and $L^\perp$ and g induces an isometry of Q; $\mathrm{Sym}(P)$ is transitive on the F_t, hence on X. So Q is a convex uniform polyhedron. □

Type (ii). The fan consists of the pyramids $\Pi_1 = (z_1; a, b, y, x)$, $\Pi_2 = (z_2; a, b, y, x)$ (the two $A_n \times \{\}$ sharing the A_n cell C, $\mathrm{VF}(C) = (a, b, y, x)$) and $T = \mathrm{conv}(a, b, z_1, z_2)$. The neighbors of v are a, b, x, y (in C) and z_1, z_2, with $z_j - v$ orthogonal to the 3-space M of C. *Cell partition:* for a vertex t let C_t be the unique A_n cell containing F_t (one A_n ray per cycle). For $t' \in C_t$, t' lies on exactly one of the two n-gons of C_t ($n \geq 4$), which is then $F_{t'}$, so $C_{t'} = C_t$. The other two neighbors t' of t satisfy $t' - t \perp M_t$, and the $A_n \times \{\}$ containing tt' gives $C_{t'} = C_t + (t' - t)$. So the vertex set is $C_0 + X$ with X in a 2-plane orthogonal to M, $P = A_n \times \mathrm{conv}\, X$, and $\mathrm{conv}\, X$ is a vertex-transitive equilateral convex polygon with unit edges, i.e. a regular polygon $\{q\}$. □

Type (iii) does not occur. The fan consists of Π_1, Π_2 as in (ii) and $T_1 = \mathrm{conv}(a, b, z_1, w)$, $T_2 = \mathrm{conv}(a, b, w, z_2)$, where $w - v$ is the lateral edge of the P_n cell between the two duoprisms. The seven points a, b, x, y, z_1, z_2, w are distinct: $a - v, b - v \in L$; $x - v$ and $y - v$ are neither in L nor orthogonal to L (x, y lie on the other n-gon of the antiprism, rotated by π/n and at height $h_A > 0$), and $x \neq y$; $w - v$, $z_1 - v$, $z_2 - v$ are orthogonal to L and are the lateral edges of three different P_n cells. Since every VF-edge lies in a fan facet, the VF-edges are exactly the edges of these four facets:

$$ab,\ ax,\ xy,\ yb,\ z_j a,\ z_j b,\ z_j x,\ z_j y\ (j = 1, 2),\ wa,\ wb,\ wz_1,\ wz_2.$$

In particular $z_1 z_2$ is not a VF-edge, and x has exactly the VF-edges xa, xy, xz_1, xz_2, y exactly yb, yx, yz_1, yz_2, and w exactly wa, wb, wz_1, wz_2; each of x, y, w is simple (Lemma 2.10). The triangle (x, y, z_1) is a 2-face of Π_1, hence a VF-ridge, lying in Π_1 and exactly one other VF-facet K'. At x, Π_1 has the edges $\{xa, xy, xz_1\}$, so K' has $\{xy, xz_1, xz_2\}$; thus xa is not an edge of K', and by Lemma 2.9 $a \notin K'$. Likewise, at y, $b \notin K'$. If w were in K', its K'-edges would be among wz_1, wz_2 (the others go to a, b), degree at most 2 in a 3-polytope, impossible. So $K' = \mathrm{conv}(x, y, z_1, z_2)$, a tetrahedron, and its edge $z_1 z_2$ is an edge of VF (Lemma 2.9, converse), which it is not. □

5.6. **Two or more large 2-faces at each vertex.** The fan of a large n-gon F at v is *paired* if one of its facets contains a second large 2-face through v: a type-(i) cycle with exactly one $q_i = m \geq 17$ (two such entries would give an angle sum of at least $2(1 - 2/17) + 1/3 > 2$), or type (ii) with $q = m \geq 17$. Otherwise it is *unpaired*: type (i) with all $q_i \leq 16$, type (ii) with $q \leq 16$, or type (iii) ($q_1, q_2 \leq 5$). In a paired fan the second large 2-face's VF-edge (the *partner edge*) is $[x_1, x_2]$ in type (i) and $[z_1, z_2]$ in type (ii).

The paired type-(i) configurations are, by the angle sum and Lemma 5.5: $(m, 4, 4)$, $(m, 3, 3, 3)$, $(m, 3, 6)$ for all $m \geq 17$; $(m, 3, 7)$ for $m \leq 41$; $(m, 3, 8)$ for $m \leq 23$; $(17, 3, 9)$; $(m, 4, 5)$ for $m \leq 19$; and $(m, 3, 3)$, $(m, 3, 4)$, $(m, 3, 5)$ are not closable.

5.6.1. *(P1): the family-polygon filter.* Let c be an unpaired type-(i) configuration (entries at most 16) around F containing a family polygon size $q \in \{7, 9, 11, \ldots, 16\}$. The q-gon Q_0 of the fan is a 2-face of P lying in the fan facet $\{n\} \times \{q\}$, so its cycle (Lemma 5.3 with q in place of n) contains $\{q\} \times \{n\}$. Type (iii) would need both duoprisms at Q_0 to have parameter at most 5, so the cycle of Q_0 is of type (i) with a configuration (n, c''), or of type (ii) with the duoprism $\{q\} \times \{n\}$. Comparing radii with $R^2 = r_n^2 + R_c^2$:

- type (ii) at Q_0: $R^2 = R_{A_q}^2 + r_n^2$, so $R_c^2 = R_{A_q}^2$, i.e. $R_c^2 - r_q^2 = h_A(q)^2/4$;
- type (i) at Q_0: $R^2 = r_q^2 + R_{(n,c'')}^2 = r_q^2 + r_n^2 + \delta_{c''}(n)$, so $R_c^2 - r_q^2 = \delta_{c''}(n)$, where (n, c'') is closable with $(1 - 2/n) + \sum_{p \in c''}(1 - 2/p) < 2$. For $n \geq 17$ the angle sum leaves $c'' \in \{(3,3), (3,4), \ldots, (3,9), (4,4), (4,5), (3,3,3)\}$ and restricts $(3,7)$ to $n \leq 41$, $(3,8)$ to $n \leq 23$, $(3,9)$ to $n = 17$ and $(4,5)$ to $n \leq 19$; closability (Lemma 5.5) removes $(3,3)$, $(3,4)$, $(3,5)$ (for $(3,5)$: $1 + \mathrm{ch}_5^2 < \mathrm{ch}_n^2$ for $n \geq 11$, so the chord triangle is obtuse), and the remaining ones are closable in these ranges.

Certified computation 5.11 (P1)**.** Of the 168 cyclic configurations with entries in 3..16 and angle sum below 2π, 137 are closable, 28 not closable and 3 degenerate. Of the 137, 83 contain a family

polygon. For 67 of these 83 there is a family polygon q in c such that $X := R_c^2 - r_q^2$ is certified different from $h_A(q)^2/4$, from $1/4$ ($c'' = (4,4)$), from the range $[0.186963, 3/16]$ of $\delta_{(3,3,3)}$, from the range $[1/3, 0.337262]$ of $\delta_{(3,6)}$, and from each of the finitely many values $\delta_{(3,7)}(n)$, $n \le 41$, $\delta_{(3,8)}(n)$, $n \le 23$, $\delta_{(3,9)}(17)$, $\delta_{(4,5)}(n)$, $n \le 19$. The 16 others are $(4,4,q)$ and $(3,3,3,q)$, $q \in \{7,9,11,\dots,16\}$. Hence the 67 configurations do not occur as fans in a convex uniform 5-polytope. The 16 survivors are genuine ($\{n\}\times P_q$ and $\{n\}\times A_q$: $\delta_{(4,4)} = 1/4 = R_{P_q}^2 - r_q^2$, and $(3,3,3,q)$ is the vertex configuration of A_q). *What is computed.* The first implementation, `caseb_famfilter.py` (Arb, 200 bits; R_c by certified bisection of f_c; closability by the sign of $f_c(c_{\max}) - 2\pi$; the three degenerate configurations by the exact right-triangle identities; the c'' list hard-coded with its n ranges, which the program checks against the angle sum). Independent checker `checks/check_caseb/p1check.py` (`mpmath.iv` at 200 bits, sympy for the exact identities) recomputes the c'' list from the angle condition with closability tested, uses closed forms for $(4,4)$, $(3,6)$, $(3,3,3)$ over all $n \ge 17$ including $n = \infty$, and gets the same 168/137/28/3/83/67 and the same 16 survivors, with no sporadic (c'', n) coincidence. The Arb implementation is a second rigorous implementation only together with the written rational angle and chord arguments above for its hard-coded c'' list and n ranges; its sampled tails are non-proof checks. Corruptions: changing $\delta_{(4,4)}$ by 10^{-30}, or using $h^2/2$ for $h^2/4$ in $R_{A_q}^2$, makes the corresponding survivors disappear (75 excluded).

5.6.2. *(W): the wedge lemma.* A point p is *admissible* for a fan if it lies on the VF sphere ($|p-v| = 1$, $|p-c| = R$), on the closed inner side of every fan facet's hyperplane, and at distance at least 1 from every fan vertex. Every VF-vertex that is not a fan vertex is admissible (VF is convex and the fan facets are VF-facets; Lemma 2.2). An *allowed* fan vertex is a fan vertex other than a, b and, for a paired fan, the partner ends. The partner ends are distinct from the other fan vertices: in type (ii) the apexes satisfy $z_j - v \perp M$ while $x - v, y - v \in M$; in type (i) the x_i are distinct lateral edges.

Certified computation 5.12 (W)**.** For each fan in the list below and every real $n \ge 17$, including the limit $n = \infty$ (and, in (c), every real $m \ge 17$ including $m = \infty$; in (d), every *integer* m in the stated range), there are no two points p, p' with $|p - p'| \ge \mathrm{ch}_{17}$, each admissible or an allowed fan vertex, other than two fan vertices. The list:

(a) type (i): the 70 closable configurations with entries at most 16 that are not excluded by (P1);
(b) type (ii), $q = 3,\dots,16$; type (iii), $(3,3)$, $(3,4)$, $(3,5)$;
(c) paired: $(m,4,4)$, $(m,3,3,3)$, $(m,3,6)$ and type (ii) with $q = m$, all $m \ge 17$ and $m = \infty$;
(d) paired, bounded, with m an integer: $(m,3,7)$, $17 \le m \le 37$; $(m,3,8)$, $17 \le m \le 22$; $(m,4,5)$, $17 \le m \le 18$, for all real $n \ge 17$; and $(m,3,7)$, $38 \le m \le 41$, $(23,3,8)$, $(17,3,9)$, $(19,4,5)$ at the integer point $n = m$ only.

Moreover, for each fan in the list, every pair of fan vertices other than e, the partner edge and the diagonals $x_i x_j$ of a type-(i) fan has distance below ch_{17} (the largest is the 16-gon chord $1.961571\dots$; in type (iii) $|z_1 - z_2| \le 1.943206$).

Scope of the two implementations. The Arb certificate tests pairs in which each point is admissible or an *allowed* fan vertex, so it omits pairs with one end in $\{a, b\}$ or at a partner end, and its fan-vertex distance bound covers pairs not involving a or b; the omitted cases are supplied by Lemma 5.4 and by the explicit lengths of Fact 5.2 (Lemma 5.14). The independent checker tests every fan vertex, including a, b and the partner ends (its test V), and every pair of fan vertices other than e and the partner edge (its test F), and certifies them. Its certification flag (`CERTIFIED`) requires test F only for the pairs of the "Moreover" clause; the type-(i) diagonals are tested in the same run and reported separately, and in the full replays none of them is long ("long type-(i) diagonals: none"). The proof does not need that supplementary result, since Lemma 5.14 excludes diagonals geometrically.

Parameter domains. The continuous parameter n (and m in (c)) is covered by compact charts that include the limits; the bounded families of (d) are generated at the 36 integer slices $(m,3,7)$,

$17 \le m \le 41$; $(m,3,8)$, $17 \le m \le 23$; $(17,3,9)$; $(m,4,5)$, $17 \le m \le 19$, and are *not* certified for non-integer m. Integer m suffices for polytopes. Both implementations enforce these 36 integer slices and the seven integer points $n = m$ (fractional or out-of-range m is refused before any arithmetic, and the checker's output records the parameter domain of every row).

The seven exceptional fans. Over the full range of n, (W) fails for $(m,3,7)$, $m = 38,\dots,41$, $(23,3,8)$, $(17,3,9)$, $(19,4,5)$, with rigorous interval witnesses at large finite n (e.g. $n = 547$ for $(41,3,7)$ and $(19,4,5)$), and also for the (P1)-excluded configurations $(3,11,13)$, $(4,7,9)$, $(5,6,7)$. For the seven paired fans, the partner m-gon's own cycle restricts n:

Lemma 5.13 (Partner-radius filter)**.** *If the fan of F (an n-gon) is paired of type (i) with configuration (m,c_1), c_1 one of $(3,7)$, $(3,8)$, $(3,9)$, $(4,5)$, then the fan of the partner m-gon is of type (i) with configuration (n,c_2) and $\delta_{c_1}(m) = \delta_{c_2}(n)$, or of type (ii) with $\delta_{c_1}(m) = h_A(m)^2/4$. For the seven exceptional fans this forces $c_2 = c_1$ and $n = m$.*

Proof. From F: $R^2 = r_n^2 + R^2_{(m,c_1)} = r_n^2 + r_m^2 + \delta_{c_1}(m)$. The partner's cycle contains $\{m\} \times \{n\}$, so it is of type (i) with a configuration (n,c_2) or of type (ii) with $q = n$ (type (iii) needs parameters at most 5); this gives $R^2 = r_m^2 + r_n^2 + \delta_{c_2}(n)$ or $R^2 = R^2_{A_m} + r_n^2 = r_m^2 + h_A(m)^2/4 + r_n^2$. For the seven fans $\delta_{c_1}(m) > 99$ (from 99.67 for $(17,3,9)$ to 1040.7 for $(41,3,7)$), while $h_A^2/4 < 1/4$, $\delta_{(4,4)} = 1/4$, $\delta_{(3,3,3)} < 3/16$, $\delta_{(3,6)} \le 0.3373$, so c_2 is one of the bounded families, and a certified comparison of the finitely many values (`checks/check_caseb/partner.py` and `wpoint.py`) leaves only $c_2 = c_1$, $n = m$. □

What is computed. The first implementation (`caseb_wedge`, Arb): with $\kappa = 1/(2R)$ and $\hat{c} = c/|c|$ (coordinates with $v = 0$), a VF-vertex is $p = \kappa\hat{c} + \sqrt{1-\kappa^2}\sum_j s_j f_j$ with $s \in S^3$ and (f_j) an orthonormal basis of $\hat{c}^\perp$; then $|p - p'| \ge L$ iff $|s+s'|^2 \le 4 - L^2/r^2$, the separation constraints are linear in s, and each fan facet is a linear inequality. A dual-tree branch-and-bound over dyadic s-boxes in $[-1,1]^4$ discards box pairs on which $\min|s+s'|^2 > 4 - L^2/r^2$ is certified, or a constraint certainly fails, and tests allowed fan vertices against single boxes. Parameters ($\theta = \pi/n \in [0,\pi/17]$; for unbounded paired families polar parameters (β,ρ) continuous at $n = m = \infty$; the facet inequalities of these families are rescaled by the vanishing factor so that they do not degenerate at $m = \infty$) are split until certified. Type (iii) roots ϕ are enclosed for every θ-box by interval bisection and Newton steps. Result (228 fans): 184 `CERTIFIED`, 31 `NOT CLOSABLE`, 3 `DEGENERATE`, 10 `FAILED` (the seven and the three above, each with a floating witness above ch_{17}); the seven certify at $n = m$ (`wpoint.py`). Independent checker `checks/check_caseb/` (own outward-rounded binary64 interval arithmetic, constants from `mpmath.iv` at 200 bits; a different chart: $\xi \in \mathbb{R}^3$ for the $L^\perp$ part, p_1 from the VF equation, $p_2 = \pm\sqrt{\cdot}$ on two branches; tests V (no long edge from an admissible point to any fan vertex, including a, b and the partner ends), P (no long admissible pair, via $\sum_i (p_i + p_i')^2 > 2 - 2\cos(2\pi/17)$), F (fan-vertex pairs); a piecewise linear chart for the unbounded families; type (iii) parametrized by $\cos(\phi/2) \in [2^{-20}, 1 - 2^{-40}]$ with n recovered from it, the two omitted ends excluded by certified inequalities, without root finding): 120 fans certified over the full range, the same 7 failing (with rigorous interval witnesses at integer n), all 7 certified at $n = m$; the 67 (P1)-excluded configurations: 64 certified, $(3,11,13)$, $(4,7,9)$, $(5,6,7)$ fail with rigorous witnesses. The verdicts agree fan by fan between the two implementations. Corruptions: in the checker, dropping the separation constraints makes $(4,4,7)$ fail; in the Arb implementation, dropping them makes $(4,6,10)$ fail, lowering the threshold below a fan's float maximum makes it fail, and dropping the fan-facet constraints makes type (ii) $q = 5$ fail. The interval searches of both implementations are rigorous; the independent checker is the one that tests every fan vertex directly.

Completeness of computed results. The Arb implementation requires every listed squared fan-vertex distance to be strictly below the threshold in Arb before certifying (equality or uncertainty splits the box or fails), and converts Arb endpoints to binary64 in the outward direction at every precision;

the independent summary requires an explicit inventory (full, supplementary or quick), rejects duplicates, omissions, extra fans, wrong domains and missing or pending V/P/F tests, and accepts an exceptional fan's point $n = m$ only with the partner filter's singleton recomputed. Corruption tests exercise these requirements. The full replay checks 228 Arb records (184 `CERTIFIED`, 31 `NOT CLOSABLE`, 3 `DEGENERATE`, 10 expected `FAILED`) and its seven points; independent 127 full-domain fans (120 certified, 7 expected failures), seven certified points, 64 of 67 supplementary fans, 1,642 required V/P tests all passed. So both implementations prove the whole of Computation 5.12, including the "Moreover" clause; the pairs with an end at a, b or a partner end are still left by the Arb implementation to Lemma 5.4 and Fact 5.2.

The chart box of the unbounded families. In the checker's chart for the unbounded paired families, with $s = \sin(\pi/n)$, $\sigma = \sin(\pi/m)$, $\tau = \max(s, \sigma)$, $\lambda = \min(1, \vartheta)$, $\omega = \min(1, 2 - \vartheta)$, $s = \lambda\tau$, $\sigma = \omega\tau$ $(0 \le \vartheta \le 2)$, the free coordinate Π is searched in $[-1.5, 1.5]$. This box contains every admissible point: the rescaled VF equation is $p_1 = \lambda(\tau(1 - 2\gamma_0 \cdot f) - \Pi)$, and a unit point has $|p_1|, |f| \le 1$; for $\vartheta \le 1$, $|\Pi| = |\xi_e| \le 1$; for $\vartheta \ge 1$, $\lambda = 1$ and $|\Pi| \le 1 + \tau(1 + 2|\gamma_0|) < 7514/5375 < 1.41$, using $|\gamma_0| < 25/43$ for all four families and $\tau \le \sin(\pi/17) < 23/125$. No fan-facet or separation constraint is used, and the bound holds at the endpoint charts $\vartheta = 0, 2$ and $\tau = 0$ without division by zero. The certificate rejects four corrupted certificates, and an explicit admissible point with $\Pi = 33/32$ shows that narrowing the box to $[-1, 1]$ would lose coverage. The live family construction checks the bound before using the box.

Lemma 5.14. *Let F be a large 2-face at v whose fan is in the list of Computation 5.12 (with the actual n, m of P). Then the only long VF-edges with an end among the fan vertices or an admissible point are e and, for a paired fan, the partner edge. In particular every long VF-edge is e or the partner edge.*

Proof. Let $[p, p']$ be a long VF-edge other than e and the partner edge. Its ends are VF-vertices, each a fan vertex or admissible. Neither end is a, b or a partner end, by Lemma 5.4 (those are already ends of a long edge). So both are admissible or allowed fan vertices, and Computation 5.12 leaves only the case that both are fan vertices. Pairs of fan vertices other than diagonals are shorter than ch_{17} (Computation 5.12, last part; pairs with a or b have the explicit lengths $\sqrt{2}$, 1 or $\sqrt{1 + \mathrm{ch}_n} < \sqrt{3}$ of Fact 5.2). A diagonal $x_i x_j$ of a type-(i) fan is not a VF-edge: the x_i satisfy $(x_i - v) \perp L$, $|x_i - v| = 1$ and lie on the VF hyperplane, so they lie on a circle; the fan edges $x_i x_{i+1}$ are VF-edges, so the circle order is the fan order (otherwise two fan edges would cross in their relative interiors, and a face containing the crossing point would contain all four ends); for $k \ge 4$ a diagonal crosses another diagonal in their relative interiors, so a face containing it would contain four vertices; for $k = 3$ there are no diagonals. Since every VF-vertex is a fan vertex or admissible, the last claim follows. □

Remark 5.15. The Arb certificate of (W) does not test pairs in which one point is a, b or a partner end, and Lemma 5.4 supplies exactly those cases. The independent checker tests them directly (test V over all fan vertices) and certifies them, so (W) as used here holds with or without the lemma.

5.6.3. *The reduction to two paired 2-faces.*

Proposition 5.16. *If $s \ge 2$, then $s = 2$ and the two large 2-faces F_1 (n-gon) and F_2 (m-gon) at v lie in a common facet $D = \{n\} \times \{m\}$.*

Proof. Let F be a large 2-face at v. If its fan is unpaired: a type-(i) configuration excluded by (P1) cannot occur (Computation 5.11); every other unpaired fan is in the list of Computation 5.12, and Lemma 5.14 shows that e is the only long VF-edge, so $s = 1$, a contradiction. So every large 2-face at v has a paired fan. By Lemma 5.13 the paired fan of F is in the list of Computation 5.12 at the actual n, m, and Lemma 5.14 shows that the long VF-edges are exactly e and the partner edge. So $s = 2$, and the two large 2-faces lie in the fan facet $\{n\} \times \{m\}$ that makes the fan paired. □

The paired type-(i) configurations not in the list of Computation 5.12, $(m,3,3)$, $(m,3,4)$, $(m,3,5)$, are not closable. The closability criterion of Lemma 5.5 excludes $(3,5)$: a spherical triangle of rays exists for $(n,3,5)$, $n<30$, but the center lies outside it. The paired families are covered by Computation 5.12(d) and Lemma 5.13.

5.6.4. *The paired theorem.*

Theorem 5.17. *If every vertex lies in exactly two large 2-faces F_1 (n-gon) and F_2 (m-gon), and they lie in a common facet $D=\{n\}\times\{m\}$, then P is $\{n\}\times\{m\}\times\{\}$, $A_m\times\{n\}$ or $A_n\times\{m\}$.*

Proof. Let $e_1=[a,b]$ and $e_2=[c,d]$ be the VF-edges of F_1, F_2; then $\mathrm{VF}(D)=\mathrm{conv}(a,b,c,d)$. By Corollary 5.9 every VF-edge lies in $\mathrm{fan}(e_1)$ or $\mathrm{fan}(e_2)$, so every VF-vertex is a vertex of one of the two fans. The fan of F_1 is of type (i) with configuration $(m,q_2,\dots,q_k)$, $x_1=c$, $x_2=d$, or of type (ii) with apexes $\{z_1,z_2\}=\{c,d\}$ (as $T=\mathrm{VF}(D)$); likewise for F_2 with n. The angle sum allows only $k=3$, or $k=4$ with configuration $(m,3,3,3)$ (the configurations $(m,4,p,p')$ and $(m,3,p,p')\neq(m,3,3,3)$ exceed 2π).

Simple-vertex step. A vertex z of $\mathrm{fan}(e_1)$ not in $\mathrm{fan}(e_2)$ has all its VF-edges in $\mathrm{fan}(e_1)$. If it has degree 4, each 3-subset of its edges spans a VF-facet (Lemma 2.10), and a VF-facet containing c and d contains the edge e_2 (Lemma 2.9), hence belongs to $\mathrm{fan}(e_2)$.

(i)/(i), $k=3$. $z=x_3$ has the VF-edges za,zb,zc,zd in $\mathrm{fan}(e_1)$. If $z\notin\mathrm{fan}(e_2)$, the facet spanned by $\{za,zc,zd\}$ contains e_2, so $z\in\mathrm{fan}(e_2)$ after all. In a type-(i) fan of e_2 every vertex other than c,d is at distance $\sqrt{2}$ from c and d (Fact 5.2), so $\mathrm{ch}_{q_2}=|z-d|=\sqrt{2}=|z-c|=\mathrm{ch}_{q_3}$: the configuration is $(m,4,4)$. The fan of F_2 is not of type (ii) (case (i)/(ii) with $k=3$, below) and not of type (i) with $k=4$ (case (i)/(i), $k=4$, below, with the roles of F_1 and F_2 exchanged), so by the same argument it is $(n,4,4)$, with the same third vertex z. *Translate argument:* for a vertex t let D_t be the facet containing its two large 2-faces. For $t'\in D_t$, both large 2-faces of D_t at t' are large 2-faces of P, so $D_{t'}=D_t$. The $\{n\}\times\{4\}$ and $\{m\}\times\{4\}$ facets at the edge vz contain $F_1+\tau$ and $F_2+\tau$, $\tau=z-v\perp\operatorname{span}D$, so $D_z=D+\tau$. Hence the vertex set is D_0+X with X on a line, every point of X is a vertex, so $|X|=2$ and $P=\{n\}\times\{m\}\times\{\}$.

(i)/(i), $k=4$: both fans of type (i), that of F_1 with configuration $(m,3,3,3)$. Here $|x_3-d|=|x_4-c|=\mathrm{ch}_3=1\neq\sqrt{2}$, so neither x_3 nor x_4 lies in a type-(i) fan of e_2. Then both are simple with all edges in $\mathrm{fan}(e_1)$: x_3 has x_3a,x_3b,x_3d,x_3x_4 and x_4 has x_4a,x_4b,x_4x_3,x_4c. The facet G at x_3 spanned by $\{x_3a,x_3d,x_3x_4\}$ contains x_4 and a, hence the edge x_4a (Lemma 2.9); it does not contain b (else x_3b would be a fourth edge of G at the simple vertex x_3). So at x_4 its edges are $\{x_4a,x_4x_3,x_4c\}$, and G contains c and d, so $G\in\mathrm{fan}(e_2)$ and $x_3\in\mathrm{fan}(e_2)$, a contradiction. Impossible.

(i)/(ii). $\mathrm{fan}(e_2)$ is the pair of $A_m\times\{\}$ with apexes a,b and trapezoid (c,d,y',x'), with $|dy'|=|y'x'|=|x'c|=1$ and diagonals $\sqrt{1+\mathrm{ch}_m}\in[1.7222,\sqrt{3})$, strictly between ch_5 and ch_6, hence not a chord ch_q of any polygon. If $k=3$, x_3 must lie in $\mathrm{fan}(e_2)$ (as above) at distances ch_{q_2}, ch_{q_3} from d, c; but the vertices of $\mathrm{fan}(e_2)$ other than a,b,c,d are x',y', each at distance 1 from one of c,d and at the diagonal distance from the other, and the diagonal is not a chord; impossible. If $k=4$, $(m,3,3,3)$: by the argument of the previous case one of x_3,x_4 lies in $\mathrm{fan}(e_2)$; by symmetry say x_3; $|x_3-d|=1$ forces $x_3=y'$. If $x_4\notin\mathrm{fan}(e_2)$, x_4 is simple and its facet spanned by $\{x_4a,x_4x_3,x_4c\}$ meets the pyramid $\Pi(a;c,d,y',x')$ in a face containing a,c,y'; since c,y' are opposite vertices of the base, no proper face of the pyramid contains these three, so the facet is the pyramid and $x_4\in\mathrm{fan}(e_2)$. Hence $\{x_3,x_4\}=\{y',x'\}$ and the neighbors of v are those of $\mathrm{fan}(e_2)$. *Cell partition:* at every vertex exactly one large 2-face has a type-(ii) fan (symmetries preserve fan types; this also holds when $n=m$); let C_t be its A_m cell. For $t'\in C_t$ the m-gon of C_t through t' has C_t in its cycle, so it is not of type (i): it is the type-(ii) 2-face at t', and $C_{t'}=C_t$. The other neighbors a,b of v are the apexes of $\mathrm{fan}(e_2)$, so $a-v$ and $b-v$ are orthogonal to the 3-space of C_v and $C_a=C_v+(a-v)$. So $P=A_m\times\mathrm{conv}\,X$ with $\mathrm{conv}\,X$ a regular polygon, here F_1: $P=A_m\times\{n\}$. The case (ii)/(i) gives $A_n\times\{m\}$.

(ii)/(ii). The trapezoid vertex x of $\mathrm{fan}(e_1)$ (edges xa, xy, xc, xd in $\mathrm{fan}(e_1)$, the apexes being c, d) must lie in $\mathrm{fan}(e_2)$ (otherwise the facet spanned by $\{xy, xc, xd\}$ contains e_2), so $x \in \{x', y'\}$ at distance 1 or $\sqrt{1+\mathrm{ch}_m}$ from c. But $|x-c| = \sqrt{2}$, since c is an apex of a prism over the A_n cell containing x. Impossible. □

5.7. **Proof of Theorem 1.3.** Let P have a large 2-face and let s be the number of large 2-faces at a vertex. If $s = 1$, Theorem 5.10 gives $\{n\} \times Q$ or $A_n \times \{q\} = \{q\} \times A_n$. If $s \geq 2$, Proposition 5.16 gives $s = 2$ with both large 2-faces in one duoprism, and Theorem 5.17 gives $\{n\} \times \{m\} \times \{\}$, $\{n\} \times A_m$ or $\{m\} \times A_n$. □

5.8. **Proof of the Main Theorem.** A convex uniform 5-polytope either has all 2-faces with at most 16 sides (Theorem 1.2: one of the 628 polytopes of $\mathcal{K}_{16}$, all of type (K1)–(K3)) or has a 2-face with at least 17 sides (Theorem 1.3: of type (K3)). □

6. Verification and reproducibility

6.1. **Arithmetic and software.** The first implementations use ball arithmetic (Arb through python-flint [14]) at 200 bits unless stated, and exact algebra in $\mathbb{Q}$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{5})$ (hand-written fields and SymPy minimal polynomials). In the proof routes (Section 6.2), every decision that cuts a branch, excludes a case or identifies two objects is either a certified inequality between enclosures, an exact algebraic identity, an exact equivalence certified by an explicit permutation, a combinatorial fact, or an equality of two vertices of one (possibly hypothetical) polytope decided by Corollary 2.3(a); an undecided case stops the computation or is reported `OPEN`, and the final runs report zero such cases. Floating-point arithmetic is used to propose (orderings, candidate correspondences, witnesses); the one remaining decision made from a double is covered by the proved coordinate lemma of Section 4.5.3. The `checks/check_u3/` geometric and transcendental routines and `checks/check_caseb/` use `mpmath.iv`, exact arithmetic or outward-rounded binary64 intervals. Those routines avoid Arb, but the complete replay also calls exact model and root-audit code using Arb. Both `checks/independent-s1-b2/` and `checks/independent-s23/` use Arb and SymPy; their independence is in implementation, not arithmetic library. All public stages use the same Python environment, specified in `checks/requirements.txt`.

Exact angle decisions versus enclosures. Angles are compared exactly through algebraic cosines, positive sines and phase products, rather than by equating decimal arccosines. In stage 1, flat angle sums $\sum \alpha_i = 2\pi$ and degenerate closure roots use the exact operands of Computation 3.6 and Lemma 3.4: the non-snub squared cosines lie in $\mathbb{Q}$, $\mathbb{Q}(\sqrt{2})$ or $\mathbb{Q}(\sqrt{5})$, with positive square roots and, where needed, further algebraic number-field extensions fixed by the constructions. The phase identity $\prod_j(\cos\alpha_j + i\sin\alpha_j) = 1$ is exact; a certified interval for the sum in $(\pi, 3\pi)$ fixes its winding. In B2, polygon angle-sum inequalities $\sum(1-2/q_i) < 2$ are rational comparisons, and the three right-triangle degeneracies are identities in $\mathbb{Q}(\sqrt{5})$ (`b2check.py`, `angle_ok`, `exact_degenerate`; Lemma 5.5). General angles, strict sign/branch bounds, closure-root isolation, radius separation and continuous parameter ranges use outward-rounded Arb balls or certified intervals instead (Computations 3.3, 4.1, 4.9, 5.8, 5.11, 5.12). An interval containing zero never proves an algebraic equality; exact residuals and sign information supply those decisions.

6.2. **Computational evidence and shared dependencies.** The computations consist of exact model checks, the facet-library and stage-1 cycle checks, radius separation, the finite stage-2/3 searches, and Case B's fan and parameter-domain checks. The packaged commands and inputs are described in the bundle's `README.md`. Their paths are relative to the bundle root.

Shared inputs. Independently written programs do not give fully independent proofs of every premise. The exact-exclusion checks share the exact constructions used by the independent stage-1 library. The stage-2/3 implementations share stage-1 class intervals, facet and known-name lists, exceptional-model certificates, the separation lemma and other written geometry. The alias table and construction theorems are also shared. The implementations use common arithmetic libraries, including Arb and SymPy. P1's second implementation depends on the written angle, chord and monotonicity arguments for its finite lists and tails.
Written mathematical premises. Facts 1.4 and 1.5 are cited classification theorems, not re-proved here. Wythoff and product constructions, the geometric reductions, coverage and interpretation of the finite search, and the unbounded-parameter reductions use the written arguments in this paper. A successful executable check establishes its stated finite or interval claims; it does not replace these arguments or establish priority.

6.3. **The four-dimensional control.** A bounded control using the shared interval root solver and certified stage-2/3 engines with exact four-dimensional facet adapters completed 639 classes at family cutoff 14. It recovered all 152 known objects (185 construction names), including the grand antiprism and snub 24-cell, and refuted 33 other realizations with no unresolved, missing or extra object. All 1,476 support witnesses passed the local checker; deletion and support corruptions were rejected. This is a bounded non-proof control, separate from the public replay; it does not re-prove Möller's all-parameter classification.

6.4. **How to reproduce.** First follow the Python environment setup in the bundle's `checks/README.md`, creating the single `.venv` environment. The checks require Python 3.11 or later and Bash 4.4 or later. From the top of the bundle, run

```
PYTHON="$PWD/.venv/bin/python" bash checks/verify-public.sh full
```

Only `full` mode checks all computational claims used in the theorem; `quick` mode does not check the theorem. Full mode includes P=16 stages 2/3, all W fans, certificate authentication and corruption controls. Neither mode verifies the written reductions automatically.

7. What is checked in Lean

The directory `lean/` is a Lean project with dependencies specified in `lean/lean-toolchain` and `lean/lake-manifest.json`. It contains geometric lemmas and a conditional classification of genuine convex unit-edge uniform five-dimensional polytopes. The formalization is incomplete: the external numerical data and their geometric correspondence have not been jointly instantiated and discharged in Lean.
The conditional composition. With explicit hypotheses as parameters, `lean/Uniform5/CompositionMainClassification.lean` proves

$$\forall P : \mathrm{Set}(\mathrm{Space}\,5), \quad \mathrm{UnitUniform}(5, P) \implies K_1(P) \vee K_2(P) \vee K_3(P).$$

Here `UnitUniform` means a finite convex hull of full affine dimension five, ambient vertex transitivity, recursively uniform facets and unit exposed edges. The conclusions are the explicit Wythoff orbit hulls and two product families, up to intrinsic Euclidean congruence. The geometry-equivalence module connects the hull and vertex-set formulations. The composition supplies the finite fan realization and admissibility needed to apply W and combines the small-face and large-face arguments; it does not assume a Case A or Case B conclusion in place of that composition.
Assumed inputs. Both `mainClassification` and `classification_of_core_inputs` take twelve computational hypotheses and the two cited classifications, besides their raw data. The unused FILTER, RIDGES, QFOLLOW and LU hypotheses and their data parameters have been removed. Appendix A prints the twelve declarations verbatim.

Every computational input remains an assumed bound parameter in Lean. The external computations and written arguments establish the intended numerical and geometric assertions; no full

external numerical search is discharged by the kernel. Checking report fields does not establish their geometric meaning or construct exact placements. Their correspondence to certificates, including coverage, canonical rigidity, refutation and forcing, remains an external semantic obligation.

The contracts distinguish coverage from existence. S1 asserts coverage and statuses without a merged-interval-width bound. S2 and U3 cover every actual figure or star, without asserting that every retained encoding is realizable. DEV is conditional on the hypothetical polytope's root, center and entire incident-facet star; it excludes that actual star, not every star having the same unlabeled figure. W is restricted to finite parameters, sufficient for actual polygonal faces, with real/integer ranges and exceptional slices explicit. The known models have separate existence arguments; the extra candidate needs no existence assertion.

Build and axiom audit. From `lean/`, run `lake exe cache get`, `lake build` and `lake env lean Audit.lean`. The audit rejects dependencies other than `propext`, `Classical.choice` and `Quot.sound`. These checks prove the stated implications, not their computational premises. The supplied modules express the sphere and separation lemmas, vertex-figure geometry, the centroid argument, the face lemma, rooted rigidity, local-to-global propagation, and conditional Case B arguments.

What remains outside Lean. The following written-to-computation links are not formalized: the numerical library/model correspondence (Section 3.2); stage-1 enumeration and root coverage (Section 4.1); the coverage of actual vertex figures by stage-2 signatures (Section 4.2); stage-3 encodings, identification and neighbor forcing (Section 4.4 and Proposition 4.14); the exhaustive stuck-development refutation of the extra candidate (Section 4.4.3); and the numerical B2, P1, partner-radius and W bounds used by the Case B structural proof (Section 5). The conditional geometric lemmas do not establish these hypotheses for the supplied certificates. In particular, rooted determinism (Lemma 4.13), congruence from discrete data (Lemma 4.10) and coset comparison (Lemma 4.11) are not formalized end to end for the concrete program encodings and candidate lists. The finite catalog coverage itself is derived conditionally from the cited classifications in `unitUniform4_smallFaces_catalogue`. No Lean theorem jointly instantiates the raw data and all twelve computational hypotheses with the two cited classifications; their joint truth rests on these written arguments and the external checkers. A kernel proof of their implication alone proves neither their truth nor their joint satisfiability. The classification is not a whole-theorem Lean proof.

Appendix A. The twelve Lean core hypotheses

The following declarations are copied verbatim from the shipped Lean sources. Names are in `Uniform5.Composition`, except B2, P1 and PartnerRadiusFilter in `Uniform5.CaseB`, and W in `Uniform5.Composition.Wedge`. Referenced types and predicates are in the cited source files; these declarations are assumptions, not verified instances of the certificates. P1 includes finite data as well as proof fields.

```
-- Core hypotheses: verbatim declarations from the accompanying Lean sources.

-- Uniform5.Composition.LIB
-- Source: lean/Uniform5/CompositionLibraryBalls.lean
structure LIB (D : LibraryNumericalData) : Prop where
  model_named : ∀ n, FacetLibraryModel n.val (D.model n)
  model_hull : ∀ n, D.model n = convexHull ℝ (D.vertexSet n : Set (Space 4))
  vertices_exact : ∀ n, (D.vertexSet n : Set (Space 4)) = extremePoints ℝ (D.model n)
  radius_enclosed : ∀ n, ∃ r, HasIntrinsicRadius (D.model n) r ∧
    (D.radiusBall n).Contains r
  coordinates_enclosed : ∀ n, ∀ v ∈ D.vertexSet n, ∀ j,
    (D.coordinateBall n v j).Contains (v j)
  angles_enclosed : ∀ n (hK : FinitePolytope (D.model n)) (k : ℕ)
    (G A B : Set (Space 4)) (hG : PolygonFace (D.model n) G k)
    (hA : Facet (D.model n) A) (hB : Facet (D.model n) B),
    G ⊆ A → G ⊆ B → A ≠ B → ∀ a b,
```

```
    Congruent A (cellTypeModel a) → Congruent B (cellTypeModel b) →
    (D.angleBall n k a b).Contains (geometricPolygonAngle hK hG hA hB)

-- Uniform5.Composition.S1
-- Source: lean/Uniform5/CompositionStageOneInputs.lean
structure S1 (N : LibraryNumericalData) (D : StageOneNumericalData) : Prop where
  cycle_count : D.listed.card = 14965
  listed_admissible : ∀ W ∈ D.listed, W.Admissible N
  distinct_classes : ∀ U ∈ D.listed, ∀ V ∈ D.listed,
    U.classCode = V.classCode → U = V
  complete : ∀ W : LibraryCycleWord, W.Admissible N → ∃ V ∈ D.listed, W.classCode = V.classCode
  reciprocal_enclosed : ∀ W ∈ D.listed,
    (D.reciprocalBall W).Contains (1 / W.maxRadius N)
  roots_cover : ∀ W ∈ D.listed, ∀ s ∈ Icc 0 (D.reciprocalBall W).upper,
    W.Domain N s → W.closure N s = 0 → ∃ I ∈ D.roots W, s ∈ I.carrier
  with_roots : D.count (fun W => D.roots W ≠ []) = 14428
  without_roots : D.count (fun W => D.roots W = []) = 537
  flat_exact : ∀ W ∈ D.listed, D.flat W = true ↔ W.angleSum N = 2 * Real.pi
  flat_count : D.count (fun W => D.flat W = true) = 7080
  degenerate_exact : ∀ W ∈ D.listed, ∀ i,
    D.degenerate W i = true ↔ D.EndpointInterval N W i
  degenerate_count : D.degenerateCount = 57

-- Uniform5.Composition.S2
-- Source: lean/Uniform5/CompositionStageTwoInputs.lean
structure S2 (N : LibraryNumericalData) (S : StageOneNumericalData)
    (T : StageTwoNumericalData S) : Prop where
  signature_count : stageTwoSignatureCount N S T = 629
  complete : ∀ c : ActiveRootClass S, c.val ∈ survivingRootClasses N S →
    ∀ R ∈ radiusClassRegion S c, ∀ D (hD : StageTwoFigureConditions N S c.val R D),
      D.signature hD ∈ T.signatures c
  empty_class : ∃ c : ActiveRootClass S, c.val ∈ survivingRootClasses N S ∧
    T.signatures c = ∅ ∧
    (∀ d : ActiveRootClass S, d.val ∈ survivingRootClasses N S → T.signatures d = ∅ → d = c) ∧
    ∃ W ∈ rootClassCycles S c.val, emptyStageTwoClassWord W

-- Uniform5.Composition.ROOT
-- Source: lean/Uniform5/CompositionRootInput.lean
structure ROOT (N : LibraryNumericalData) (S : StageOneNumericalData)
    (T : StageTwoNumericalData S) : Prop where
  edge_root : ∀ c, c.val ∈ survivingRootClasses N S →
    ∀ (σ : VertexFigureSignature), σ ∈ T.signatures c →
    ∃ W ∈ σ.edges,
      W.closure N (reciprocalClassHull S c).lo *
        W.closure N (reciprocalClassHull S c).hi < 0 ∧
      (∀ s ∈ (reciprocalClassHull S c).carrier, deriv (W.closure N) s ≠ 0) ∧
      ∃! s, s ∈ (reciprocalClassHull S c).carrier ∧ W.closure N s = 0

-- Uniform5.Composition.DEV
-- Source: lean/Uniform5/CompositionU3DevelopmentInput.lean
structure DEV (N : LibraryNumericalData) (S : StageOneNumericalData)
    (U : U3RealizationData S) (extra : U3StarData) : Prop where
  refutation : ∀ X, U3GeometricStar N S (U.rowClass u3RefutedRow).val X →
    X.radius ∈ radiusClassRegion S (U.rowClass u3RefutedRow) →
    (U.rowSignature u3RefutedRow).RadiusCompatible N X.radius →
    U.canonicalEncoding X = U.rowEncoding u3RefutedRow →
    ∀ P : UnitUniform5, ∀ v ∈ P.val.extremePoints ℝ,
      X.root = v → X.center = polytopeCenter P →
      X.star N = facetStar P.val v → False

-- Uniform5.CaseB.B2
-- Source: lean/Uniform5/CaseBInputs.lean
structure B2 (Ω : Set ℝ) : Prop where
  type_i : ∀ (n : ℕ), 17 ≤ n → ∀ c : List ℕ,
    AngleCondition (realConfig c) → Closable (realConfig c) →
```

```
    Real.sqrt (polygonRadius n ^ 2 + configRadius (realConfig c) ^ 2) ∉ Ω
  type_ii : ∀ (n q : ℕ), 17 ≤ n → 3 ≤ q →
    Real.sqrt (antiprismRadiusSq n + polygonRadius q ^ 2) ∉ Ω
  type_iii : ∀ (n q₁ q₂ : ℕ), 17 ≤ n → TypeIIIPair q₁ q₂ →
    ∀ γ : ℝ, -1 < γ → γ < 1 → TypeIIIPolynomial n q₁ q₂ γ = 0 →
    Real.sqrt (antiprismRadiusSq n + 1 / (2 * (1 + γ))) ∉ Ω

-- Uniform5.CaseB.P1
-- Source: lean/Uniform5/CaseBInputs.lean
structure P1 where
  configurations : Finset (List ℕ)
  closable : Finset (List ℕ)
  nonclosable : Finset (List ℕ)
  degenerate : Finset (List ℕ)
  family : Finset (List ℕ)
  excluded : Finset (List ℕ)
  configurations_iff : ∀ c, c ∈ configurations ↔ SmallConfiguration c
  closable_iff : ∀ c, c ∈ closable ↔ c ∈ configurations ∧ Closable (realConfig c)
  nonclosable_iff : ∀ c, c ∈ nonclosable ↔ c ∈ configurations ∧
    configArcSum (realConfig c) (configMax (realConfig c)) < 2 * Real.pi
  degenerate_iff : ∀ c, c ∈ degenerate ↔ c ∈ configurations ∧
    configArcSum (realConfig c) (configMax (realConfig c)) = 2 * Real.pi
  family_iff : ∀ c, c ∈ family ↔ c ∈ closable ∧ ∃ q ∈ c, FamilyPolygon q
  excluded_subset : excluded ⊆ family
  configurations_count : configurations.card = 168
  closable_count : closable.card = 137
  nonclosable_count : nonclosable.card = 28
  degenerate_count : degenerate.card = 3
  family_count : family.card = 83
  excluded_count : excluded.card = 67
  separation : ∀ c ∈ excluded, ∃ q ∈ c, FamilyPolygon q ∧ P1Separated c q
  delta44 : ∀ n : ℕ, 17 ≤ n → delta [4, 4] n = 1 / 4
  delta333_range : ∀ n : ℕ, 17 ≤ n →
    delta [3, 3, 3] n ∈ Set.Icc (186963 / 1000000 : ℝ) (3 / 16)
  delta36_range : ∀ n : ℕ, 17 ≤ n →
    delta [3, 6] n ∈ Set.Icc (1 / 3 : ℝ) (337262 / 1000000)
  survivors : ∀ c, c ∈ family ∧ c ∉ excluded ↔
    ∃ q, FamilyPolygon q ∧ (c = [4, 4, q] ∨ c = [3, 3, 3, q])

-- Uniform5.Composition.Wedge.W
-- Source: lean/Uniform5/CompositionWedgeInput.lean
structure W (p1 : CaseB.P1) : Prop where
  no_long_pair : ∀ n f, CaseB.LargeParameter n → CaseB.WListed p1 n f →
    ∀ F, RealizesWFan p1 n f F → ∀ p q,
    (F.Admissible p ∨ F.Allowed p) → (F.Admissible q ∨ F.Allowed q) →
    CaseB.chord 17 ≤ dist p q → p ∈ F.vertices ∧ q ∈ F.vertices
  fan_distances : ∀ n f, CaseB.LargeParameter n → CaseB.WListed p1 n f →
    ∀ F, RealizesWFan p1 n f F → ∀ p ∈ F.vertices, ∀ q ∈ F.vertices,
    ¬ F.specialPair p q → dist p q < CaseB.chord 17

-- Uniform5.CaseB.PartnerRadiusFilter
-- Source: lean/Uniform5/CaseBInputs.lean
structure PartnerRadiusFilter : Prop where
  exclude_ii : ∀ m c₁, ExceptionalPaired m c₁ →
    delta c₁ m ≠ antiprismHeightSq m / 4
  force_equal : ∀ (n m : ℕ) (c₁ c₂ : List ℕ), 17 ≤ n → ExceptionalPaired m c₁ →
    PartnerTail n c₂ →
    AngleCondition ((n : ℝ) :: realConfig c₂) → Closable ((n : ℝ) :: realConfig c₂) →
    delta c₁ m = delta c₂ n → c₂ = c₁ ∧ n = m

-- Uniform5.Composition.U3Realizations
-- Source: lean/Uniform5/CompositionU3RealizationInputs.lean
structure U3Realizations (N : LibraryNumericalData) (S : StageOneNumericalData)
    (T : StageTwoNumericalData S) (U : U3RealizationData S) : Prop where
  class_valid : ∀ r ∈ u3ReportP16,
```

```
    (U.rowClass r).val ∈ survivingRootClasses N S
  class_indices : ∀ r ∈ u3ReportP16, ∀ s ∈ u3ReportP16,
    U.rowClass r = U.rowClass s → r.classIndex = s.classIndex
  signature_valid : ∀ r ∈ u3ReportP16,
    U.rowSignature r ∈ T.signatures (U.rowClass r)
  signature_complete : ∀ c : ActiveRootClass S,
    c.val ∈ survivingRootClasses N S → ∀ σ ∈ T.signatures c,
      ∃ r ∈ u3ReportP16, U.rowClass r = c ∧ U.rowSignature r = σ
  signature_indices : ∀ r ∈ u3ReportP16, ∀ s ∈ u3ReportP16,
    U.rowClass r = U.rowClass s → U.rowSignature r = U.rowSignature s →
      r.signatureIndex = s.signatureIndex
  complete : ∀ (c : ActiveRootClass S), c.val ∈ survivingRootClasses N S →
    ∀ X (hX : U3GeometricStar N S c.val X), X.radius ∈ radiusClassRegion S c →
      X.figure.signature hX.figure ∈ T.signatures c →
      ∃ r ∈ u3ReportP16, U.rowClass r = c ∧
        U.rowSignature r = X.figure.signature hX.figure ∧
        U.canonicalEncoding X = U.rowEncoding r
  encoding_rigid : ∀ (c : ActiveRootClass S), c.val ∈ survivingRootClasses N S →
    ∀ X Y, U3GeometricStar N S c.val X → U3GeometricStar N S c.val Y →
      X.radius ∈ radiusClassRegion S c → Y.radius ∈ radiusClassRegion S c →
      X.radius = Y.radius → U.canonicalEncoding X = U.canonicalEncoding Y →
        U3StarCongruent N X Y

-- Uniform5.Composition.U3Identification
-- Source: lean/Uniform5/CompositionU3IdentificationInputs.lean
structure U3Identification (N : LibraryNumericalData) (S : StageOneNumericalData)
    (U : U3RealizationData S) (M : U3KnownModelData) : Prop where
  admitted : ∀ r ∈ u3ReportP16, ∀ n : KnownFiveName,
    n.sourceName ∈ r.known_names →
      KnownFiveModel n (M.model n) ∧
      ∃ hQ : UnitUniform 5 (M.model n),
        M.root n ∈ (M.model n).extremePoints ℝ ∧
        M.center n = vertexCentroid (vertices (M.model n) (uniform_finitePolytope hQ.1)) ∧
        intrinsicRadius (M.model n) ∈ radiusClassRegion S (U.rowClass r) ∧
        ∀ F, Facet (M.model n) F →
          ∃ s ∈ u3ReportP16, ∃ k ∈ (U.rowSignature s).facets, Congruent F (N.model k)
  soundNO : ∀ r ∈ u3ReportP16, ∀ n : KnownFiveName,
    (n.sourceName, U3EmbeddingVerdict.no) ∈ r.embedding →
      ∀ X, U3GeometricStar N S (U.rowClass r).val X →
        X.radius ∈ radiusClassRegion S (U.rowClass r) →
        U.canonicalEncoding X = U.rowEncoding r → ¬ U3MatchesKnown N M n X
  soundNOCOUNT : ∀ r ∈ u3ReportP16, ∀ n : KnownFiveName,
    (n.sourceName, U3EmbeddingVerdict.noCount) ∈ r.embedding →
      ∀ X, U3GeometricStar N S (U.rowClass r).val X →
        X.radius ∈ radiusClassRegion S (U.rowClass r) →
        U.canonicalEncoding X = U.rowEncoding r → ¬ U3MatchesKnown N M n X

-- Uniform5.Composition.U3Forcing
-- Source: lean/Uniform5/CompositionU3ForcingInputs.lean
structure U3Forcing (N : LibraryNumericalData) (S : StageOneNumericalData)
    (U : U3RealizationData S) (D : U3ForcingData) : Prop where
  checked : ∀ r ∈ u3ReportP16, r.local_uniqueness.forced = true →
    ∀ X, U3GeometricStar N S (U.rowClass r).val X →
      X.radius ∈ radiusClassRegion S (U.rowClass r) →
      U.canonicalEncoding X = U.rowEncoding r →
      ∀ w, starNeighbor X.root w (X.star N) → U3ForcingCheck N D X w
```

Appendix B. Names

A name such as `B40001` lists the rings on nodes $0, 1, 2, 3$ (here the tesseract; `B41000` is the 16-cell). Cell labels: `tet`, `oct`, `cube`, `ico`, `doe` (Platonic); `tut`, `co`, `tic`, `toe`, `sirco`, `girco`, `id`, `tid`, `ti`, `srid`, `grid`, `snic`, `snid` (truncated tetrahedron, cuboctahedron, truncated cube, truncated octahedron, rhombicuboctahedron, truncated cuboctahedron, icosidodecahedron, truncated icosahedron,

TABLE 2. Coxeter diagrams and node orders used in the names.

Group	Diagram (nodes $0, 1, \dots$)
A_4, A_5	linear, all branches 3
B_4	linear $3, 3, 4$; the 4 between nodes 2 and 3
B_5	linear $3, 3, 3, 4$; the 4 between nodes 3 and 4
D_4	node 1 joined to nodes 0, 2, 3 (all 3)
D_5	node 2 joined to nodes 1, 3, 4; node 0 joined to node 1
F_4	linear $3, 4, 3$; the 4 between nodes 1 and 2
H_4	linear $5, 3, 3$; the 5 between nodes 0 and 1

truncated dodecahedron, rhombicosidodecahedron, truncated icosidodecahedron, snub cube, snub dodecahedron, in this order; note that `tid` is the truncated icosahedron and `ti` the truncated dodecahedron, the reverse of Bowers' usual names); `P`n the n-gonal prism ($n \neq 4$) and `A`n the n-gonal antiprism ($n \neq 3$).

TABLE 3. The 33 aliases of the facet library and their canonical names (Computation 3.2).

Canonical	Aliases	Canonical	Aliases
`A40001`	`A41000`	`B41000`	`D40001`, `D40010`, `D41000`
`A40010`	`A40100`	`B41010`	`D41011`, `F40010`, `F40100`
`A40011`	`A41100`	`B41100`	`D40101`, `D40110`, `D41100`
`A40101`	`A41010`	`B41110`	`D41111`, `F40011`, `F41100`
`A40111`	`A41110`	`F40101`	`F41010`
`A41011`	`A41101`	`F40111`	`F41110`
`B40001`	`pr-cube`, `duoprism-4-4`	`F41011`	`F41101`
`B40010`	`D40011`, `D41001`, `D41010`	`pr-oct`	`antiprism-prism-3`
`B40100`	`D40100`, `F40001`, `F41000`		
`B40110`	`D40111`, `D41101`, `D41110`		

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